

# INTRAHOUSEHOLD INEQUALITY AND THE JOINT TAXATION OF HOUSEHOLD EARNINGS\*

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## Abstract

We derive optimal labor income tax schedules for married agents, taking the distinction between interpersonal and interhousehold inequality seriously. Each household consists of two workers with different productivity levels and unequal access to the family's economic resources. We handle the multidimensionality that could undermine the [Mirrlees' \(1971\)](#) approach by restricting preferences to be identical and iso-elastic and by focusing on taxes characterized by income-splitting. After showing how individual-oriented utilitarianism typically leads to a misalignment between the households' and the government's objectives, which [Apps and Rees \(1988\)](#) have named *dissonance*, we provide a complete solution for the screening problem, incorporating different degrees of assortative matching and assessing dissonance's role in shaping the optimal schedule. We also investigate the welfare gains from gender-based policies. **Keywords:** *Intrahousehold Inequality; Joint Taxation; Collective Household; Multidimensional Screening.* **JEL Classification:** H21, H31, D13.

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RESOURCES are not equally split across household members, and this fact has significant and sometimes dramatic consequences for interpersonal inequality and poverty. For instance, recent work has shown that inequality within the household generates higher poverty figures that amount to higher mortality rates between gender. For example, [Calvi \(2020\)](#) finds that, in India, the share of resources allocated to women within households starts declining after the age of 45. Moreover, [Anderson and Ray \(2010\)](#) report that women start dying at increasing rates in India precisely at the same age. Given these figures, ignoring intra-household inequality is a significant drawback for properly assessing inequality and welfare distribution, which is the objective of redistribution policies.<sup>1</sup>

The main goal of this paper is to think about optimal taxation of families, with an emphasis on its consequence for interpersonal rather than inter-household inequality. We focus on how to use the tax system to redistribute resources within a household. In order to do so, we consider families in which the allocation of economic resources and well-being is not equitable, following the Collective approach of household behavior as in [Chiappori \(1988b, 1992\)](#); [Apps and Rees \(1988\)](#). We also assume that the government cares about individuals directly and not the household; however, it cannot directly redistribute within the household. Therefore, our optimal tax system uses redistribution across households to affect the distribution of resources within households.

Households in our model are composed of two spouses, each with a utility function and an individual level of labor market productivity. Resources are allocated across spouses according to their relative power, captured by household specific Pareto weights. Since the primitive object of concern for economic policies is the well-being of individuals, tax policy is assessed according to its consequence for each person, instead of household. Welfare analysis is, therefore, revised following the change in paradigm, from Unitary to Collective, that took place in family economics.

We start our analysis by completely characterizing the optimal joint income tax system for the case of identical iso-elastic preferences. To do so, we must deal with the screening problem described by [Mirrlees \(1971\)](#), but in a multidimensional context. Indeed, household tax theory has always been limited by the multidimensional nature of the screening problem to which it gives rise. As a consequence, most contributions to the literature limit its scope by pre-defining functional forms for taxes ([Boskin and Sheshinski \(1983\)](#)), precluding adjustments in all of the relevant decision margins ([Kleven et al. \(2009\)](#)), and/or restricting the amount of heterogeneity ([Cremer et al. \(2012\)](#)).

Our first contribution is to offer an aggregation result to reduce the dimen-

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<sup>1</sup>See [Chavas et al. \(2018\)](#); [Chiappori and Meghir \(2015\)](#); [Haddad and Kanbur \(1990\)](#).

sionality of the screening problem and enable the use of standard tools to solve the mechanism design problem. As it turns out, under the assumption of identical iso-elastic preferences, we are able to maintain full generality on skill distributions, the relative Pareto weight of spouses, and the correlation between spouses' skills while still solving the model. Here, the inherent multidimensionality that plagues optimal household taxation endogenously collapses into a single-dimensional problem in which single-crossing still holds. Since there is only one screening variable (household earnings) and three dimensions of characteristics (spouses' productivities and the spouses' relative weight parameter), at each earning level there is a bunching of different household types. However, because of the specification of preferences, these sets are identified by a common parameter.<sup>2</sup> This substantially simplifies the analysis and allows us to derive a formula for the optimal marginal tax rate under general distributions of household types. The optimal tax formula, which is written in terms of primitives of the model, is explicit about how family decision-making affects the shape of the optimal tax system and can be successfully taken to the data.

Households assess their own well-being using a Bergsonian household utility function involving different Pareto weights applied to each spouse's utility. The planner, in contrast, uses individual-oriented utilitarianism, aggregating individuals' well-being as the sum of their utility.<sup>3</sup> The potential misalignment between household and social objectives – *dissonance*, in the terminology of Apps and Rees (2011) – leads to an adjustment of the traditional Mirrlees (1971) formula. In the presence of dissonance, the traditional formula for optimal marginal taxes is augmented by a Pigouvian correction term that internalizes the impacts of taxes on the distribution of resources within the household. This new term comes into place due to the presence of dissonance and adjusts for the intrahousehold inequality. Taken as a function of household Pareto weights, this Pigouvian correction term has a U-shaped format and is equal to zero when government and household objectives are perfectly aligned; which corresponds to equal Pareto weights for both spouses. Starting from equal weights, the Pigouvian term increases (becomes pos-

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<sup>2</sup> Exogeneity of this aggregated parameter with respect to policy is key, since this is what allows for an exogenous ordering of households willingness to make effort. For general preferences, the ordering becomes endogenous to the tax system and the screening problem is not tractable. Moreover, even if we settle for a local characterization via perturbation methods, the aggregation of preferences at the same income level introduces novel elements to the analysis which we explore in Section B.

<sup>3</sup>By neglecting the multiperson nature of households, the welfare evaluation typical of optimal taxation literature is devoid of any meaning if we are committed to methodological individualism, a central tenet in the economic analysis. In fact, when coining the term "methodological individualism", Schumpeter (1954) wrote: "the self-governing individual constitutes the ultimate unit of social sciences; and that all social phenomena resolve themselves into decisions and actions of individuals that need not or cannot be further analyzed in terms of superindividual factors."

itive) as the Pareto weight is biased against the most productive spouse and decreases (becomes negative) in the opposite direction. The minimum occurs at zero only when spouses have the same productivity. In a sense, households for which the relative weight is biased toward the most productive spouse overestimate the cost of effort when compared to the planner's evaluation.

We illustrate how intrahousehold inequality affects the shape and size of taxes with an empirical application that estimates the optimal joint income tax using data from the March 2016 supplement of the Current Population Survey (CPS). In our main calibration, we find the Pigouvian correction to significantly change the shape of optimal income taxes. Our simulations also suggest that a planner concerned with the intrahousehold distribution of welfare induces more intense redistribution; the actual distribution of resources across individuals is a mean preserving spread of the distribution across representative household members. In our exercise, the planner only has joint taxes as an instrument for redistributing across households, which will be used to the extreme to achieve a more equitable outcome at the individual level.

Compared with the benchmark case without dissonance, in our baseline scenario, the planner induces an allocation characterized by a substantially – 64% – higher average income and substantially lower pre-tax dispersion in income. To understand the first aspect, note that, for very unequal couples, i.e., those for which spouses have very different productivities, the cost of increasing effort as perceived by the couple is significantly higher than how the planner values it. The planner is more willing to sacrifice leisure to increase consumption than a planner who does not take the utility of individuals into account. As for the inequality level, measured by the Gini index, it is slightly lower, even though the average marginal tax rate is higher when dissonance is *not* taken into account. This is explained by the fact that the pre-tax distribution induced by the planner with dissonance concerns is substantially more compressed.

We restrict most of our analysis to the case of joint tax schedules.<sup>4</sup> Still, in Section 5.1, we go beyond joint taxation by investigating the welfare gains of gender-based taxation policies. We show that a small tax on the earnings of the least productive spouse redistributes utility toward this spouse. The logic behind

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<sup>4</sup>Despite a growing tendency toward individual taxes, joint tax schedules are still pervasive. The usual rationale for their use is based on horizontal equity principles: one should not treat couples with identical earnings differently. Although we do not necessarily side with this view, we acknowledge the compelling argument by [Gordon and Kopczuk \(2014\)](#), who show that, from an equity perspective, joint taxes are a better departure point for tax design than individual taxes. Starting with a schedule that depends only on total household earnings (or on individual earnings in the case of individual taxation), [Gordon and Kopczuk \(2014\)](#) estimate the benefits of conditioning taxes on other observables. They show that departing from separate taxes requires more conditioning and accomplishes less in terms of promoting equity.

this result is that a spouse-specific income tax is indeed a subsidy to the spouse's leisure. Under the assumption that Pareto weights are not affected by the new policy, the distribution of consumption is not altered and the relative utility of the taxed spouse is increased. This effect must, of course, be balanced by the overall decrease in welfare as perceived by the household. Still, when the least productive spouse is also the one with less power – the case emphasized by [Immervoll et al. \(2011\)](#) – optimal policy will typically introduce a small tax on this agent.<sup>5</sup> This finding depends on the assumption adopted in almost all of the household taxation literature ([Kleven et al. \(2009\)](#); [Immervoll et al. \(2011\)](#); [Cremer et al. \(2016\)](#)) that household weights are invariant to perturbations to the policy but need not be overturned and is, in fact, more likely to be reinforced by the presence of household production. We make explicit the reliance on fixed Pareto weights by considering a very stylized collective model in which household weights depend on the relative net marginal productivity of spouses. Under this assumption, the intuitive result is recovered: taxing a spouse hurts him/her.

The assumption of identical iso-elastic preferences may leave one wondering how general our findings may be. To partially answer this question, we use a tax perturbation approach to derive optimal tax formulae in terms of sufficient statistics with no restrictions on preferences, as in [Saez \(2001\)](#). The presence of dissonance modifies the formula in two important ways. The first is the introduction of an additional sufficient statistic that should be taken into account: the wedge between social values of income and its market price. The second concerns the aggregation of elasticities across households with the same earnings. The multidimensionality of household types leads to heterogeneity in preferences at each earning level. This was already the case in [Saez \(2001\)](#) since he did not impose single-crossing. The novelty here that arises from dissonance is that there is not one but two different relevant aggregated elasticities: the usual average elasticities and a welfare-weighted elasticity.<sup>6</sup> Government revenues depend on average elasticities in which the average at each level of earning is calculated using the empirical distribution. Yet, elasticities are also important statistics because dissonance makes behavioral responses relevant to welfare.<sup>7</sup> In this case, the average elasticity is obtained by aggregating across households at each earning level using a welfare-adjusted distribution, analogously to the risk adjustment typically used

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<sup>5</sup>This finding is also in line with [Cremer et al. \(2016\)](#).

<sup>6</sup>In [Saez's](#) words, "It is not necessary to assume that people earning the same income have the same elasticity; the relevant parameters are simply the average elasticities at given income levels" ([Saez \(2001\)](#), p. 210). The complexity introduced by dissonance is the aggregation issue mentioned in footnote 2, is shown to also arise in the study of optimal taxation of behavioral agents by [Gerritsen \(2016\)](#).

<sup>7</sup>In particular, the envelope arguments used to restrict the welfare consequences of tax perturbations to the mechanical effect cannot be used if there is dissonance.

in finance.

For the types of preferences typically used in optimal taxation studies, our optimal tax formula has an ABC plus D representation, which makes it a natural extension of the ABCs formulas first derived by [Diamond \(1998\)](#).

The rest of the paper is organized as follows. After this introduction, we offer a brief literature review. Section 1 describes the economy. The policy objective is carefully discussed in Section 2. In Section 3, we show how the assumption of identical iso-elastic preferences allows us to reduce the dimensionality of the screening program to actually solve the model in terms of primitives. Section B generalizes the optimal formulae from Section 3 using a tax perturbation approach. These formulae are based on sufficient statistics that are not too easy to recover and that furthermore are endogenous to the policy. Our approach in this section highlights the type of assumptions that are needed if the approach is to be put into practice. In Section 5, we explore a simple form of gender-dependent perturbation on the joint schedule, and allow for the existence of child in the household. Section 6 concludes. Proofs are relegated to Appendix D.

## Related literature

Until recently, economists have approached household behavior on a common preference basis where all family members' resources were pooled to maximize a single objective function. Moreover, this "household welfare function" had a normative status when used in optimal tax theory (e.g., [Boskin and Sheshinski \(1983\)](#)). However, starting with the pioneering contributions of [Manser and Brown \(1980\)](#) and [McElroy and Horney \(1981\)](#) and the conceptual change promoted by [Chiapori \(1988b, 1992\)](#); [Apps and Rees \(1988\)](#), Family Economics rapidly progressed toward the understanding of joint family decisions driven by divergent interests within the household.<sup>8</sup>

[Boskin and Sheshinski \(1983\)](#) explore the sub optimality of tax schedules such as the one adopted in the US in which husbands and wives face equal marginal tax rates. They discuss the possibilities of taxing spouses equally or differently, or even subsidizing one of the spouses. They show that taxing the earnings of husbands and wives at the same rate is inefficient because it disproportionately reduces female labor market participation. [Boskin and Sheshinski \(1983\)](#) use a unitary approach and take household (revealed) preferences as the normative criterion to be used by the planner. Moreover, they restrict their analysis to linear

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<sup>8</sup>[Lundberg and Pollak \(1996\)](#), for example, adopted a bargaining protocol to model the independent agency of men and women in marriage. Early contributions to the understanding of the internal workings of families can be found in many of Gary Becker's contributions.



taxes. [Bastani \(2013\)](#) extends the analysis of linear taxes to a collective setting in which spouses decide through bargaining.

Also under a unitary approach, [Kleven et al. \(2009\)](#) analyze the general non-linear optimal income tax for couples. In their model, the primary earner chooses labor supply as a continuum (intensive margin), while the secondary earner decides whether or not to participate in the labor market (extensive margin). If the secondary earner opts to participate in the job market, the labor supply is given, whereas in our environment, both spouses choose in the intensive margin.<sup>9</sup> They show conditions under which the optimal tax scheme displays a positive tax on secondary earnings and when those taxes on secondary earnings decrease with primary earnings. [Kleven et al.](#)'s analysis starts from individual taxes and asks whether introducing jointness is optimal. Our analysis in Section ?? takes the exact opposite starting point, (i.e., joint taxes), and asks whether introducing differentiation on marginal taxes can be welfare improving.

Our assumption regarding household preferences and collective behavior implies the type of dimensionality-reduction exogenously imposed by [Choné and Laroque \(2010\)](#) while retaining the information that permits the assessment of individual welfare. A similar approach is used by [Heathcote and Tsujiyama \(2021\)](#) under the assumption of perfect insurance across large families. The distribution of power and dissonance are not considered.

Closer to our work are the studies by [Immervoll et al. \(2011\)](#), [Cremer et al. \(2012\)](#), and [Cremer et al. \(2016\)](#). Dissonance, as defined by [Apps and Rees \(1988\)](#), plays a central role in all of these works. [Immervoll et al. \(2011\)](#) simplify their analysis by only considering extensive margin decisions and imposing strong restrictions on the choice sets. [Cremer et al. \(2012\)](#) handle multidimensionality by restricting couple types with the assumption of perfect assortative matching. Finally, [Cremer et al. \(2016\)](#) consider only a finite number of types. However, they consider fully general taxes and show how tax formulae change when compared to a world without dissonance. The small number of types they consider allows them to focus on a world in which incentive constraints only bind in the usual direction.

[Guner et al. \(2012\)](#) quantify the effects of tax reforms taking carefully into account the labor supply of married women as well as the current demographic structure. Married women have a heterogeneous cost of participating. They analyze how the structure of taxation can affect the participation decision, which is empirically responsive to tax perturbations. In our setting, both men and women have a heterogeneous participation cost, and they do not have the option to file

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<sup>9</sup>In a previous version of our work, we explain how the analysis could be extended to allow for the secondary earner to adjust labor supply in both margins.

separately if both spouses are working.

Our model is static, so we refrain from discussing the important issues related to the interaction between risk sharing and spouses' labor supply analyzed in [Blundell et al. \(2016\)](#) and [Wu and Krueger \(2019\)](#). However, as in these works, joint labor supply decisions are the essence of our analysis.

Finally, it is important to mention [Gerritsen's \(2016\)](#) study of behavioral agents. By distinguishing 'decision utilities' from 'experience utilities' a misalignment between individuals' and planners' objectives akin to dissonance arises in his work. Similar concerns with aggregation play a role in his optimal tax formulae.

## 1 Environment

The economy is inhabited by a continuum of households (or families) with measure normalized to one. Each household is composed of two spouses indexed by  $i \in \{a, b\}$ , where  $a$  identifies the female spouse and  $b$  the male spouse in the household. Each spouse is characterized by a type  $w_i \in [\underline{w}, \bar{w}] \subset \mathbb{R}_+$  denoting individual labor market productivity (or wage rate). Technology is linear as one efficient unit of labor produces one unit of consumption good. Agents are paid their marginal productivity, which implies that the labor income generated by spouse  $i \in \{a, b\}$  whose productivity is  $w_i$  when that spouse supplies  $l_i$  hours of work is given by  $z_i = w_i l_i$ .

Individuals derive utility from the consumption of a private good  $x_i \in \mathbb{R}_+$  and disutility from marketable labor supply  $l_i \in \mathbb{R}_+$  resulting in a total utility given by  $U(x_i, l_i)$ , common to all individuals. Utility  $U : \mathbb{R}_+^2 \rightarrow \mathbb{R}$  is assumed to be strictly quasi-concave, strictly increasing in consumption, strictly decreasing in labor supply, and of class  $\mathcal{C}^2$ .

**Household decision making** We take the multiperson nature of households seriously. We also assume that households are collective, in the sense that household consumption and labor supply choices always result in efficient outcomes, regardless of the bargain protocol they engage in. Therefore, without loss of generality, we can assume that a household whose members have productivities  $(w_a, w_b)$  make consumption and labor supply decisions  $(x_i, l_i)_{i \in \{a, b\}}$  to maximize a Bergsonian household utility function of the form

$$\alpha U(x_a, l_a) + (1 - \alpha) U(x_b, l_b), \quad (1)$$

for some Pareto weight  $\alpha \in [0, 1]$ , reflecting the relative contribution of the female spouse to household welfare. We impose no restrictions on the relationship



between the Pareto weight,  $\alpha$ , and spouses' productivities,  $(w_a, w_b)$ .

Given the previous structure, each household can be parameterized by a triple  $(w_a, w_b, \alpha) \in \Lambda \equiv [\underline{w}, \overline{w}] \times [\underline{w}, \overline{w}] \times [0, 1]$ . This triple will henceforth be referred to as the family type, denoted by  $\iota \equiv (w_a, w_b, \alpha)$ .

**Informationally feasible allocations** We follow the household economics literature by assuming that only total household consumption  $x \equiv x_a + x_b$  is observable outside the household; that is, consumption cannot be assigned to a specific spouse. Because of this assumption, goods cannot be conditioned on each spouse's consumption.<sup>10</sup> Given this informational structure, bundles that can be associated with a household in any mechanism are of the form  $(x, z_a, z_b) \in \mathbb{R}_+^3$ .<sup>11</sup>

Moreover, in most of this paper (except for Section ??), we restrict our analysis to policies that cannot distinguish the income generated by each of the spouses. In this case, feasible bundles are of the form  $(x, z) \in \mathbb{R}_+^2$ , where  $z \equiv z_a + z_b$  is total income of the household. We call them *aggregate bundles*.

**Tax system** For most of what follows, we focus on joint taxation, which only uses information on total household income  $z = z_a + z_b$ , that is,  $\mathbb{T}(z_a, z_b) = T(z_a + z_b)$ .<sup>12</sup> Therefore, a government with a redistributive objective taxes households using a nonlinear joint income tax schedule,  $T : \mathbb{R}_+ \rightarrow \mathbb{R}$ , assigning a tax liability for each possible level of family income. Call  $\mathcal{T}_0$  the set of all such tax systems. The net of tax income is used in the family consumption  $x = z - T(z)$ .

Studying joint taxation is particularly important because of its empirical relevance.<sup>13</sup> For instance, in the US, couples have the choice of filing taxes individually or jointly. Under the joint tax option, the marginal tax rates depend exclusively on the aggregate income of the household. The overall progressivity of the labor income tax schedule means that filing individually is almost never optimal.<sup>14</sup> Al-

<sup>10</sup>A more general approach is to define assignable and non-assignable goods. For the former, the spouse who is actually consuming each quantity can be determined, whereas for the latter, this is not possible. One could also include public goods consumed by both spouses in a nonrivalrous manner. For simplicity, there is only one non-assignable private consumption good. This classification is independent of the relation between taxation and labor choices emphasized in this paper.

<sup>11</sup>Equivalently,  $(x, z_a, z_b)$  is such an allocation if and only if  $\phi(x, z_a, z_b) \leq 0$  for some function  $\phi : \mathbb{R}_+^3 \rightarrow \mathbb{R}$ . These are the only budget sets that can be designed to induce a desired allocation by a planner.

<sup>12</sup>The polar opposite is a separable tax schedule – a function  $\mathbb{T} : \mathbb{R}_+^2 \rightarrow \mathbb{R}$  that assigns to each vector of household incomes  $(z_a, z_b) \in \mathbb{R}_+^2$  a tax to be paid  $\mathbb{T}(z_a, z_b) = T(z_a) + T(z_b) \in \mathbb{R}$ , for some  $\mathcal{C}^2$  function  $T : \mathbb{R}_+ \rightarrow \mathbb{R}$ . Any deviation from this separable tax system is not neutral to marital status and may impose a marriage penalty or bonus.

<sup>13</sup>Cremer et al. (2012) provide conditions under which the government should tax couples jointly.

<sup>14</sup>Ireland and Germany are some other countries that use this type of joint taxation.

though agnostic about the foundations of joint taxation, we view fairness as an appealing property for its use.

This restriction on the space of feasible mechanisms will be convenient in reducing the number of margins that a government has to take into account in order to align incentives in this multidimensional environment.<sup>15</sup>

Following [Immervoll et al. \(2011\)](#) we assume that  $\alpha$  is exogenous and independent of policy.

**Family problem** As we have argued before, an important characteristic of the household consumption data is that most consumption is not assignable in the sense that only  $x$ , not  $x_a$  and  $x_b$ , is observed by outsiders. Moreover, under the restriction that policy instruments are based on total household earnings, the budget constraint is invariant to the source of labor income, and we can define for a type  $\iota = (w_a, w_b, \alpha)$  family a utility function  $V : \mathbb{R}_+^2 \times \Lambda \rightarrow \mathbb{R}$  as

$$V(x, z, \iota) \equiv \max_{(x_a, x_b), (z_a, z_b) \in \mathbb{R}_+^2} \left\{ \alpha U\left(x_a, \frac{z_a}{w_a}\right) + (1 - \alpha)U\left(x_b, \frac{z_b}{w_b}\right) \right. \\ \left. \text{s.t. } x_a + x_b = x \quad \text{and} \quad z_a + z_b = z \right\}. \quad (2)$$

The utility function  $V$  reflects the collective nature of family decisions and incorporates the intrahousehold allocation decisions of how much each spouse contributes to family income and how resources are distributed between family members.<sup>16</sup> It represents the ordering of  $(x, z)$  bundles by a household composed of spouses with productivities  $w_a$  and  $w_b$  and Pareto weight  $\alpha$ . From this formulation, it is easy to see that different spouses from families choosing the same  $(x, z)$  can achieve very different utility levels depending on their relative productivities and Pareto weights.

Two important assumptions are embedded in this formulation:  $\alpha$  is exoge-

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<sup>15</sup>[Mirrlees \(1971\)](#) addresses the problem of choosing a fully nonlinear tax system as a screening problem. It has long been recognized that screening problems in multidimensional environments are hard to handle because the absence of an exogenous ordering of willingness to pay or work precludes a simple characterization of the binding constraints. It is also well known that the optimum of these problems usually implies too much *pooling*; see [Rochet and Choné \(1998\)](#), and [Rochet and Stole \(1987\)](#).

<sup>16</sup>If we do not restrict the instruments beyond the informational constraint imposed by the non-assignability of consumption, the household utility for a type  $\iota$  family would be

$$\mathcal{V}(x, z_a, z_b, \iota) \equiv \max_{(x_a, x_b) \in \mathbb{R}_+^2} \left\{ \alpha U(x_a, z_a/w_a) + (1 - \alpha)U(x_b, z_b/w_b) \quad \text{s.t. } x = x_a + x_b \right\}.$$

This last definition will be useful when we consider gender-based policies in Section ?? . Of course, we can always define  $V$  in (2) through  $V(x, z, \iota) \equiv \max_{z_b \in \mathbb{R}_+} \mathcal{V}(x, z - z_b, z_b, \iota)$ .

nous to any policy, and the decision process within the family will always lead to an efficient outcome. This latter assumption places our approach in the realm of collective models (e.g., [Chiappori \(1988a\)](#) and [Browning and Chiappori \(1998\)](#)). As for the former, we could hold Pareto weights fixed for two plausible reasons. First, Pareto weights could be determined at the marriage stage, as in [Gayle and Shephard \(2019\)](#).<sup>17</sup> Second, as in [da Costa and de Lima \(2019\)](#), the planner may have additional instruments to handle Pareto weights. In both cases, an interesting question we leave for future work is how policies influence the distribution of weights in the long run through marriage markets.

Since intrahousehold transfers are not observed, the allocation of consumption between spouses cannot be directly affected by any policy. Hence, for any feasible policy, household preferences ordering can be represented by the family utility function  $V$ .

## 2 The Planner's Problem

A utilitarian planner with fully flexible tax instruments would eliminate any intrahousehold inequality by setting effort and consumption across spouses accordingly. We take on the more realistic case in which the planner has access to imperfect instruments to influence intrahousehold inequality. In fact, we start with joint tax systems  $T \in \mathcal{T}_0$  that do not discriminate between the sources of income. This section presents an optimal tax calculation to answer the question of how a government can use redistribution between households to achieve redistribution within the family. Family characteristics  $\iota$  are heterogeneous in the population and assumed to be private information of the family. Denote  $F(\iota)$  as their joint distribution over the support  $\Lambda \equiv [\underline{w}, \bar{w}] \times [\underline{w}, \bar{w}] \times [0, 1]$ . The planner can use the tax policy  $T : \mathbb{R}_+ \rightarrow \mathbb{R}$  to influence couples' aggregate decisions and spouses' allocations indirectly. We assume that the planner is restricted to using joint tax schedules  $T \in \mathcal{T}_0$ .

### 2.1 The planner's objective

We have associated with each household  $\iota$  a utility function  $V(x, z, \iota)$  representing its preferences ordering on aggregate bundles. Whether any welfare meaning should be attached to it is a different matter.

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<sup>17</sup>We do not take into account the policy impact on household formation. That is, we assume that couples do not divorce in response to changes in policies, nor do we allow spouses to anticipate those changes at the moment of marriage. [Alm and Whittington \(1999\)](#) find that in the US, the impacts of the income tax on marriage decisions, even when statistically significant, are small.

As Chiappori and Meghir (2015) have pointed out, a welfare metric that does not take into account intrahousehold inequality is inconsistent with any welfare criteria based on individuals, as is the case for utilitarianism. Traditional principles of morals and ethics do not justify the planner taking into account households', instead of individuals', utilities for welfare evaluation. Moreover, intrahousehold inequality is central to the study of redistribution policies. The goal of this paper is exactly to derive the implications of this intrahousehold inequality to the optimal tax theory.

Therefore, even though the planner takes "household preferences" as given, its objective may depend arbitrarily on the utilities of all individuals. We shall focus on an anonymous individual-oriented utilitarian objective (i.e., the planner maximizes the average of all individuals' utilities).

**Dissonance** Under the individual-oriented utilitarian objective, the planner weights equally the utility of spouses in a given couple. Spouses, in contrast, decide using a different objective (1) captured by their Pareto weights  $\alpha$  and  $1 - \alpha$ . The solution to (2) defines consumption  $x_a(x, z, \iota)$  and income  $z_a(x, z, \iota)$  choices for the wife as a selection from the argmax correspondence. We can also directly define the analogous functions  $x_b(x, z, \iota) = x - x_a(x, z, \iota)$  and  $z_b(x, z, \iota) = z - z_a(x, z, \iota)$  for the husband.

Using the notation above, the household welfare as perceived by the planner is

$$W(x, z, \iota) \equiv \frac{1}{2} \left\{ U \left( x_a(x, z, \iota), \frac{z_a(x, z, \iota)}{w_a} \right) + U \left( x_b(x, z, \iota), \frac{z_b(x, z, \iota)}{w_b} \right) \right\}. \quad (3)$$

The function,  $W : \mathbb{R}_+^2 \times \Lambda \rightarrow \mathbb{R}$ , represents the planner's assessment of individuals' welfare in a household  $\iota$  when consuming the aggregate bundle  $(x, z)$ . Let  $z_T$  denote the household choice of  $z$  under a smooth tax schedule,  $T \in \mathcal{T}_0$ . We can in this case define  $\xi$  (NEED A NAME) through

$$dW = \frac{\partial W(x, z, \iota)}{\partial x} [1 - T'(z)] + \frac{\partial W(x, z, \iota)}{\partial z} = [1 - \xi(x, z, \iota)] \frac{\partial W(x, z, \iota)}{\partial x} [1 - T'(z)], \quad (4)$$

where

$$\xi(x, z, \iota) := \frac{\partial W(x, z, \iota)/\partial z}{\partial W(x, z, \iota)/\partial x} \left[ \frac{\partial V(x, z, \iota)/\partial z}{\partial V(x, z, \iota)/\partial x} \right]^{-1}$$

is equal to one whenever  $\alpha = 1/2$ .

We can also re-write (3) as

$$W(x, z, \boldsymbol{\iota}) = V(x, z, \boldsymbol{\iota}) + \left( \frac{1}{2} - \alpha \right) \left[ U \left( x_a(x, z, \boldsymbol{\iota}), \frac{z_a(x, z, \boldsymbol{\iota})}{w_a} \right) - U \left( x_b(x, z, \boldsymbol{\iota}), \frac{z_b(x, z, \boldsymbol{\iota})}{w_b} \right) \right]. \quad (5)$$

This expression has a nice behavioral analogue.<sup>18</sup> The first term on the right hand side of (5) is the spouses' aggregate utility under the household welfare metric (2). It is analogous to the *decision utility*, of some behavioral economics models – see Farhi and Gabaix (2020). The planner's ordering of bundles,  $W$ , whose analogue is the *experience utility* of behavioral economics does not coincide with the household's ordering (except when  $\alpha = 1/2$ ). The difference between the two – the second term on the right-hand side of (5) – determines the misalignment between the household's and planner's welfare metric in the assessment of spouses' well-being.

Note that, even if the ordering represented by  $V(x, z, \boldsymbol{\iota})$  coincides with the one represented by  $W(x, z, \boldsymbol{\iota})$  (leading to the same choices  $(x, z)$ ), the value assigned by the planner to a unit of consumption good in the hand of each household may still depend on  $\alpha$ . Indeed, how  $(x, z)$  is split between spouses matters and, in the case of equal preferences and productivity, unless household resources are equally divided between the spouses, that is,  $x_a(x, z, \boldsymbol{\iota}) = x_b(x, z, \boldsymbol{\iota})$  and  $z_a(x, z, \boldsymbol{\iota}) = z_b(x, z, \boldsymbol{\iota})$ , there would be misalignment even if the household chose  $(x, z)$  exactly as the planner would.<sup>19</sup>

### 3 Optimal Taxation

By the revelation principle, the planner's problem can be restated in the space of direct mechanisms as

$$\max_{(x, z): \Lambda \rightarrow \mathbb{R}_+^2} \int_{\Lambda} W(x(\boldsymbol{\iota}), z(\boldsymbol{\iota}), \boldsymbol{\iota}) dF(\boldsymbol{\iota}),$$

subject to incentive compatibility constraints: for every  $\boldsymbol{\iota} \in \Lambda$

$$\boldsymbol{\iota} \in \arg \max_{\tilde{\boldsymbol{\iota}} \in \Lambda} V(x(\tilde{\boldsymbol{\iota}}), z(\tilde{\boldsymbol{\iota}}), \boldsymbol{\iota}),$$

---

<sup>18</sup>A couple in our model is analogous to a behavioral agent of Farhi and Gabaix (2020) and Gerritsen (2016).

<sup>19</sup>This is the case when instead of using equal weights to both spouses in (3), the planner would follow the weights  $\alpha$  to the wife and  $1 - \alpha$  to the husband as in (1).

where we slightly abused notation in using  $x(\cdot)$  and  $z(\cdot)$  to denote the outcome function associated with the direct mechanism and the resource constraint:

$$\int_{\Lambda} [z(\iota) - x(\iota)] dF(\iota) \geq 0.$$

To circumvent the technical difficulties that arise from the multidimensional nature of the screening problem, we specialize our model in this section. We consider a class of preferences for which the household type collapses into a one-dimensional type with preferences that still possess the single-crossing property. Assume separable preferences of the form  $U(x_i, l_i) = u(x_i) - h(l_i)$  with  $u(x_i) = x_i^{1-\sigma}(1-\sigma)^{-1}$ , for  $\sigma > 0$ ,  $\sigma \neq 1$ ,  $u(x_i) = \ln x_i$ , for  $\sigma = 1$ , and  $h(l_i) = l_i^{1+\gamma}(1+\gamma)^{-1}$ ,  $\gamma > 0$ . Using  $l_i = z_i/w_i$ , we can re-parameterize each agent's utility as

$$U(x_i, z_i, w_i) = \frac{x_i^{1-\sigma}}{1-\sigma} - \frac{1}{1+\gamma} \left( \frac{z_i}{w_i} \right)^{1+\gamma}, \quad (6)$$

These preferences are in line with the empirical evidence reviewed in [Chetty \(2006\)](#) and it is one of the most often parametrization for preferences used in the taxation literature (see [Tuomala \(2016\)](#)).

We shall show next is that, under an income-splitting schedule, the earnings of a type  $\iota = (\alpha, w_a, w_b)$  household can be represented as if they were made by a representative individual with preferences

$$u(x) - \theta(\iota)h(z) \quad (7)$$

where

$$\theta(\iota) := \frac{w(\iota)^{-1-\gamma}}{[\alpha^{1/\sigma} + (1-\alpha)^{1/\sigma}]^\sigma}. \quad (8)$$

and

$$w(\iota) := \left[ \alpha^{\frac{-1}{\gamma}} w_a^{\frac{1+\gamma}{\gamma}} + w_b^{\frac{1+\gamma}{\gamma}} (1-\alpha)^{\frac{-1}{\gamma}} \right]^{\frac{\gamma}{1+\gamma}}, \quad (9)$$

is the 'household productivity'.

Two features are worth noting in this household utility function. First, family types only enter the objective through  $\theta(\iota)$ . Therefore, families whose characteristics generate the same  $\theta(\iota)$  share the same objective. We can think of  $\theta(\iota)$  as a sufficient statistic for the family characteristics when computing the utility of a given aggregate bundle  $(x, z) \in \mathbb{R}_{++} \times \mathbb{R}_+$ . Second, marginal rates of substitution  $\theta(\iota)h'(z)/u'(c)$  are exogenously ordered and we can follow Mirrlees' approach screening households by their unitary type  $\theta(\iota)$ .

In other words, the combination of iso-elastic preferences for spouses and joint income-splitting taxes allows us to collapse the household's multi-dimensional



type  $\iota$  into a uni-dimensional aggregator that preserves single-crossing. Assuming the existence of such aggregator allows Choné and Laroque (2010) to study taxation in a multi-dimensional setting. Here we show that, as in Heathcote et al. (2017), household cooperative behavior implies the existence of such an aggregator.

For every  $\theta$ , it will be convenient to define the set  $\mathcal{I}(\theta) \equiv \{\iota \in \Lambda | \theta(\iota) = \theta\}$  and the  $\theta$ -household ‘wedge’,

$$\bar{\xi}(\theta) \equiv \mathbb{E} [\xi(x, y, \iota) | \iota \in \mathcal{I}(\theta)] .$$

While  $\xi(x, y, \iota)$  may depend on  $(x, y)$ , under the specification of preferences we are using, it does not.

We can split the household decision problem into two stages: (i) the allocation of consumption and effort across spouses and (ii) the optimal choice of total consumption and effort given the tax system.

To see that, consider a household’s least costly way of producing  $z$ , that is

$$\mathcal{H}(z|\iota) = \min_{z_a} \left\{ \frac{\alpha}{1+\gamma} \left( \frac{z_a}{w_a} \right)^{1+\gamma} + (1-\alpha) \left( \frac{z-z_a}{w_b} \right)^{1+\gamma} \right\} \quad (10)$$

It is easy to check that spouses’ contribution to total household income can be expressed in terms of productivities,  $\theta_a, \theta_b$ , and the Pareto coefficient  $\alpha$  as

$$\frac{z_a}{z} = \underbrace{(1-\alpha)^{\frac{1}{\gamma}} \frac{w_a^{\frac{1+\gamma}{\gamma}}}{w(\iota)^{\frac{1+\gamma}{\gamma}}}}_{k_a} \quad \text{and} \quad \frac{z_b}{z} = \underbrace{\alpha^{\frac{1}{\gamma}} \frac{w_b^{\frac{1+\gamma}{\gamma}}}{w(\iota)^{\frac{1+\gamma}{\gamma}}}}_{k_b} . \quad (11)$$

Substituting  $z_a, z_b$  in (10) gives us

$$\mathcal{H}(z|\iota) = \frac{1}{1+\gamma} \left( \frac{z}{w(\iota)} \right)^{1+\gamma} . \quad (12)$$

A spouse’s contribution to household income varies negatively with the spouse’s own individual Pareto weight; not only consumption but also leisure increases with one’s Pareto weight. Individual earnings also vary positively with the spousal productivity gap, which means that more productive spouses work relatively more, holding all other primitives fixed.

Now let’s consider the allocation of consumption between spouses. A simple

application of Borch's rule lead to

$$\mathcal{U}(x|\boldsymbol{\iota}) = \max_{x_a} \left\{ \alpha \frac{x_a^{1-\sigma}}{1-\sigma} + (1-\alpha) \frac{(x-x_a)^{1-\sigma}}{1-\sigma} \right\} = B(\alpha) \frac{x^{1-\sigma}}{1-\sigma}, \quad (13)$$

where  $B(\alpha)$  is the denominator in the right hand side of (8).

Combining both facts, we have

$$V(x, z, \boldsymbol{\iota}) = \mathcal{U}(x|\boldsymbol{\iota}) - \mathcal{H}(z|\boldsymbol{\iota}) = B(\alpha) \left[ \frac{x^{1-\sigma}}{1-\sigma} - \theta(\boldsymbol{\iota}) \frac{z^{1+\gamma}}{1+\gamma} \right], \quad (14)$$

for  $\sigma \neq 1$  and

$$V(x, z, \boldsymbol{\iota}) = \mathcal{U}(x|\boldsymbol{\iota}) - \mathcal{H}(z|\boldsymbol{\iota}) = \kappa(\alpha) + \ln x - \theta(\boldsymbol{\iota}) \frac{z^{1+\gamma}}{1+\gamma},$$

where  $\kappa(\alpha) \equiv \alpha \ln \alpha + (1-\alpha) \ln(1-\alpha)$  for  $\sigma = 1$ .

In all that follows we focus on the case  $\sigma = 1$ . This will simplify many of our expressions and allow us to obtain closed form expressions for some of the extensions we study. The counterpart for the dissonance expressions we derive are trivial generalizations of the ones we use here. The only feature that specializing for  $\ln$  preferences eliminates is the possible dependence of marginal social welfare weights on  $\alpha$ .<sup>20</sup> While interesting *per se*, the possibility that Pareto weights vary with aspects of an agent type has already been study in other contexts, e.g., Choné and Laroque (2010), so we skip it for brevity.

Before we continue with our analysis, a brief discussion of our assumptions is needed. First, because we assume that men and women have the same preferences, particularly the same Frisch elasticity, they would have identical labor supply elasticities if unmarried. Since our focus is on married persons, however, labor supply responds to both changes in one's own productivity as well as her spouse's productivity. It is not hard to check that for realistic values for the household's elasticity of taxable income, the cross-sectional elasticity of taxable income with respect to one's own productivity is larger for the secondary earner than for the primary earner. Indeed, it is always the case if preferences are strongly separable.

Second, the particular parameterization of preferences we choose implies that both spouses always supply a positive quantity of labor.<sup>21</sup> Hence, decisions are taken along the intensive margin only. In a previous version of the paper – Alves

<sup>20</sup>The marginal social value of one unit of consumption in the hands of a couple with net income  $x$  and household weight  $\alpha$  is  $0.5 \left[ (\alpha)^{1/\sigma} + (1-\alpha)^{1/\sigma} \right]^\sigma x^{1-\sigma}/(1-\sigma)$  for  $\sigma \neq 1$  and  $\ln x$  for  $\sigma = 1$ .

<sup>21</sup>The choice of  $\ln$  parameterization implies that the utility is not defined under null consumption. This way, the family utility will not be well defined when either  $\alpha = 0$  or  $\alpha = 1$ .

et al. (2021) – we discuss participation decisions in an extension of the model where an additional discrete participation cost is introduced in the model. These two aspects allow us to address better the observed differences in labor supply elasticities of married men and women.

Given our assumptions, the problem of a type  $\theta$  household facing a tax schedule  $T : \mathbb{R}_+ \rightarrow \mathbb{R}$  – i.e., stage (ii) – is given by

$$\max_{(x,z) \in \mathbb{R}_{++} \times \mathbb{R}_+} \left\{ \ln x - \theta \frac{z^{1+\gamma}}{1+\gamma}, \quad \text{s.t.} \quad x = z - T(z) \right\}. \quad (15)$$

Again, we want to stress that this problem is shared by all families whose combinations of  $\iota = (w_a, w_b, \alpha)$  generate the same  $\theta$ , i.e.,  $\iota \in \mathcal{I}(\theta)$ . This representation allows us to think about the planner's program as if it were a program involving a representative individual for the family whose type is the one-dimensional parameter  $\theta$  and derives utility from the aggregate bundle  $(x, z)$ . Aggregate preferences are well behaved and exogenously ordered with respect to the parameter  $\theta$  (i.e., the *Spence-Mirrlees condition* is satisfied). The first-order necessary condition for the type  $\theta$  family's problem (15) is

$$1 - T'(z) = x\theta z^\gamma.$$

### 3.1 Optimal tax formulae

In the rest of this section, let us consider a government that taxes households using a nonlinear joint income tax schedule  $T : \mathbb{R}_+ \rightarrow \mathbb{R}$  to maximize a utilitarian social welfare function.

To write the planner's problem define, for all  $\theta$

$$\bar{W}(x, z, \theta) \equiv [W(x, y, \iota) | \iota \in \mathcal{I}(\theta)], \quad \text{and} \quad \bar{V}(x, z, \theta) \equiv [V(x, y, \iota) | \iota \in \mathcal{I}(\theta)],$$

and note that

$$\bar{W}(x, z, \theta) = \bar{V}(x, z, \theta) + [1 - \bar{\xi}(\theta)] \theta \frac{z^{1+\gamma}}{1+\gamma}.$$

This allows us to write the planners' objective as

$$\max_{(x(\theta), z(\theta))_\theta} \int_{\underline{\theta}}^{\bar{\theta}} \left\{ \bar{V}(x(\theta), z(\theta), \theta) + [1 - \bar{\xi}(\theta)] \theta \frac{z^{1+\gamma}}{1+\gamma} \right\} \psi(\theta) d\theta, \quad (16)$$

while keeping the incentive compatibility constraint,

$$\theta \in \operatorname{argmax}_r V(x(r), z(r), \theta),$$

and the resource constraint,

$$\int_{\underline{\theta}}^{\bar{\theta}} [z(\theta) - x(\theta)] \psi(\theta) d\theta \geq G,$$

identical to the ones in the single persons households' **Mirrlees'** program.

**Proposition 1.** *In all intervals where  $z(\theta)$  is strictly increasing, the marginal tax rate for the optimal (nonlinear) income joint taxation is*

$$\frac{T'(z(\theta))}{1 - T'(z(\theta))} = \frac{1}{\psi(\theta)x(\theta)\theta} \int_{\underline{\theta}}^{\bar{\theta}} [x(s) - \mathbb{E}^\psi[x]] \psi(s) ds - \frac{\mathbb{E}^\psi[x]}{x(\theta)} [1 - \bar{\xi}(\theta)]. \quad (17)$$

*Proof.* See Appendix D. □

### 3.2 A Pigouvian Connection

As we will try to convey next, traditional optimal tax formulae must be adjusted to include a Pigouvian term that accounts for the differential impact of changes in redistribution of utility across spouses. To understand this effect we write the difference in utility attained by spouses as

$$V_a(z - T(z), z, \iota) - V_b(z - T(z), z, \iota) = \ln \frac{\alpha}{1 - \alpha} - \left[ \left( \frac{k_a}{w_a} \right)^{1+\gamma} - \left( \frac{k_b}{w_b} \right)^{1+\gamma} \right] \frac{z^{1+\gamma}}{1 + \gamma},$$

where  $k_i$ ,  $i = a, b$ , is the share of  $z$  earned by spouse  $i$  – see (11).

A slight change in  $z$  has a different utility impact on the two spouses whenever  $k_a/w_a \neq k_b/w_b$ . In particular if  $k_a/w_a > k_b/w_b$  then increasing  $z$  will increase more (decrease less) the utility of spouse  $b$  than that of spouse  $a$ .

**The Utilitarian Consequence** From the Utilitarian planner's perspective the impact of a small change in  $a$ ,  $dz$ , is

$$dV_a + dV_b = \left\{ \frac{1 - T'(z)}{z - T(z)} - \frac{1}{2} \left[ \left( \frac{k_a}{w_a} \right)^{1+\gamma} + \left( \frac{k_b}{w_b} \right)^{1+\gamma} \right] z^\gamma \right\} dz$$

which, using the household's optimality condition, can be written<sup>22</sup>

$$\begin{aligned} dV_a + dV_b &= \left\{ \left[ \alpha \left( \frac{k_a}{w_a} \right)^{1+\gamma} + (1-\alpha) \left( \frac{k_b}{w_b} \right)^{1+\gamma} \right] - \frac{1}{2} \left[ \left( \frac{k_a}{w_a} \right)^{1+\gamma} + \left( \frac{k_b}{w_b} \right)^{1+\gamma} \right] \right\} z^\gamma dz \\ &= [1 - \xi(x, z, \iota)] \left[ \alpha \left( \frac{k_a}{w_a} \right)^{1+\gamma} + (1-\alpha) \left( \frac{k_b}{w_b} \right)^{1+\gamma} \right] z^\gamma dz. \end{aligned}$$

It is not hard to see that  $\xi(x, z, \iota) \leq 1 \Leftrightarrow k_a/w_a \leq k_b/w_b$ , thus providing the conditions under which it will be optimal to distort taxes relative to the Mirrlees' optimum.

## 4 Quantitative Assessment

We now compute the marginal tax rate derived in Proposition 1 and Corollary ?? using real data from the March 2016 extract of the CPS. This exercise will provide a quantitative assessment of how family decision making affects the size and shape of optimal joint income taxes. In Appendix E, we have a detailed exposition of the implementation based on a discrete grid of the income distribution adapted from [Mankiw et al. \(2009\)](#) to our model.

We restrict our sample to married spouses whose incomes are strictly positive. As in [Mankiw et al. \(2009\)](#), we use hourly wages as a proxy for productivity. The parameter  $\gamma$  in the disutility of labor is set to be equal to  $\gamma = 1.5$ , implying a Frisch elasticity of labor supply of 0.66. It is important to note that the effects uncovered in our numeric assessment are heavily dependent on the degree of assortative matching. In these simulations, we implicitly use the degree inherent to our sample, which is equal to 0.2129 when measured by the correlation between spouses' hourly wages.

For this policy evaluation, we postulate that all families share the same Pareto weight,  $\alpha$ . This simplification helps us in the implementation and interpretation of the forces behind the optimal taxes. By fixing the degree of intrahousehold inequality, we are able to isolate its impact on tax prescriptions. Under this assumption, the marginal tax rate in Corollary ?? depends on the distributions of the family unitary type  $\theta$ , its distribution  $\psi(\theta)$ , as well as the distribution of the modified type  $\theta_a$  conditional on  $\theta$ ,  $\phi(\theta_a | \theta)$ . We calculate these distributions from the knowledge of the empirical wage distribution. In our baseline scenario, we use  $\alpha = 0.46$ . This choice of  $\alpha$  generates the average ratio of leisure consumption

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<sup>22</sup>When preferences are separable  $\xi(x, z, \iota)$  does not depend on  $x$ . For the special case of identical iso-elastic preferences one can readily see that nor it depends on  $z$ .

between spouses that we observe in the data.<sup>23</sup>

Table 1: Baseline Calibration

| Parameter | Value | Source-Target            |
|-----------|-------|--------------------------|
| $\gamma$  | 1.5   | Frisch Elasticity = 0.66 |
| $\sigma$  | 1     | $u(x) = \ln x$           |
| $\alpha$  | 0.46  | Same relative Leisure    |

Recall the formula for the optimal marginal tax rate in Corollary ?? . The last term on the right-hand side of this formula is the additional distortion that the planner imposes to influence the intra-household decision process. It depends on the determinants of collective decision-making  $\alpha$  and the conditional expectation  $\mathbb{E}[\theta_a|\theta]$  as well as the determinants of behavioral responses.

Our baseline calculation for the optimal marginal tax rate is displayed in Figure 1. Note that in the horizontal axis, we use the family productivity  $\theta(\iota)^{-(1+\gamma)}$  to facilitate interpretation. We show the relevance of taking dissonance into account by both displaying the marginal tax rate when the planner attaches normative meaning to household choices, i.e., ignores dissonance, and by decomposing the optimal tax rate into the typical Mirrleesian term and the dissonance term. That is, the graph displays four curves. In solid green is the optimal marginal tax rate, as in Corollary ??, where the planner maximizes individual-oriented utilitarian criteria. We decompose this optimal marginal tax in two terms. The first term on the right-hand side in (??), which we refer to as the standard term, is the dashed pink line with diamond markers, and the second term on the right-hand side in (??), which we refer to as the dissonance term, is the point-wise blue line with square markers. Lastly, the solid red line is the optimal marginal tax rate, as in (??), where the planner maximizes household-oriented utilitarian criteria.

Although  $\alpha = 0.46$  is very close to the household-oriented utilitarian objective, 0.5, the Pigouvian correction term has dramatic consequences for optimal taxes.<sup>24</sup> The reason is that the dissonance crucially depends on the ratio between spouses productivities which in our data set displays some very large values for this ratio, which is not surprising since the correlation between spouses' productivities is only 0.2129 in our data set. Note also that, although the standard term (dashed

<sup>23</sup>This number is very close to the estimated bargaining weights in Lise and Yamada (2018) using Japanese panel data.

<sup>24</sup>In the appendix we consider scenarios in which the planner maximizes individual-oriented and household-oriented utilitarian criteria for different, even if counterfactual, assumption about the household Pareto weights. These explorations bring to the forefront what is the relevant policy object in the equity-efficiency trade-off that constitutes the core of the optimal taxation problem.

pink line with diamond markers) results from the same expression of optimal marginal tax without dissonance, its contribution to the optimal tax with dissonance is not the same. This happens because, the two are evaluated at different allocations.

[ FIGURE 1 ABOUT HERE ]

When dissonance is taken into account, marginal taxes are lower than when they are not. At first, this seems counter-intuitive: if the government wants more distribution, why are marginal taxes not higher? The government aims to induce less pre-tax income inequality by reducing the overall progressivity of the system.

Indeed, the pre-tax earnings for the least productive family are substantially higher when the planner takes dissonance into account. Moreover, less progressivity leads to a smaller cross-sectional elasticity of taxable income, translating into a less dispersed income distribution.

Table 2: Gini Coefficients Implied by the Optimal Taxation  $\alpha = 0.46$

|                    | Household Level | Individual Level |
|--------------------|-----------------|------------------|
| Without Dissonance | 0.1574          | 0.1639           |
| With Dissonance    | 0.0969          | 0.1071           |

The reported gini coefficient refers to after tax income.

Also, important is the difference in inequality when measured at the individual and the household level. Table 2 displays pre- and after-tax inequality measured by the Gini coefficient at the household and individual levels. Whenever preferences are separable and the utility of consumption is identical for the two spouses, interpersonal inequality exceeds inter-household inequality. Heuristically, under the former criterion, one considers that spouses with low power in poor households have substantially smaller incomes than spouses with high power in rich households. Under the latter, it is only the average consumption of spouses in these two households that matters.

[ FIGURE 2 ABOUT HERE ]

This is not the whole story. While we have focused on inequality, what ultimately matters for the planner is everyone's well-being. Hence, another salient difference between the allocations that arise under the two policies is the average income produced. When dissonance is taken into account, the policy induces a substantially higher, 64.4% - higher, average pre-tax income, thus showing that



dissonance changes in a significant way how the planner values the trade-off between household consumption and total household effort.

[ FIGURE 3 ABOUT HERE ]

All these features are summarized in Figures 2 and 3 where we plot the cumulative distributions for pre and after-tax income for the two different government objectives. The stark difference between the distributions for the two objectives highlights the two features aforementioned. First, the levels of income are higher for almost all households under dissonance. It is only at the top of the distribution that dissonance seems to motivate the discouragement of effort. This is expected; we mostly find couples for which both spouses are highly productive at the very top. Second, with dissonance, redistribution relies less on the mapping from pre- to after-tax income and more on not allowing for a very dispersed pre-tax distribution of income.

## 5 Extensions

The study of Kleven et al. (2009) is perhaps the main reference for the optimal taxation of couples under a mechanism design approach. The complexities related to the multidimensional nature of the problem are circumvented by assuming that primary earners make choices along the intensive margin and secondary earners along the extensive margin only, and by focusing on individual taxation. By introducing the notion of jointness in tax schedules, they explore the usefulness of making the marginal tax rates of a spouse who chooses in the intensive margin to depend on the earnings of the other.

In this paper, we start from the opposite perspective: tax schedules characterized by income splitting, for which marginal tax rates are always equalized across spouses. In this section we introduce a small linear marginal tax,  $t$ , on the gross income of spouse  $a$  at all income levels. The nonlinear income schedule then applies to total household income after this tax. Note how this reform has a heterogeneous impact on couples with identical  $z$  depending on how earnings are distributed across spouses.<sup>25</sup>

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<sup>25</sup>When the budget constraint distinguishes the source of income, one must make explicit the impact of a reform on each different source of taxable income (i.e.,  $z_a$  or  $z_b$ ). Yet, because the baseline schedule is of the form  $\mathcal{T}(z_a, z_b) = T(z_a + z_b)$ , the behavioral impact of any reform on tax revenues depends only on  $dz = dz_a + dz_b$ , but not on  $dz_a$  or  $dz_b$  separately. If we let  $1 = (1, 1)$ , then  $dz = 1 \cdot (dz_a, dz_b)$ . Revenue impacts are captured by the same general expressions derived in Section B.

## 5.1 Combining joint and individual instruments

Assume that we want to improve all spouse  $a$ 's situation by offering them a linear earnings' subsidy,  $t \in \mathbb{R}$ . In this case the household's budget constraint becomes

$$x_a + x_b \leq z_a(1 + t) + z_b - T(z_a(1 + t) + z_b).$$

Let  $z = z_a(1 + t) + z_b$ , then, we can write the families effort allocation problem as

$$\min_{z_a} \alpha \left( \frac{z_a}{w_a} \right)^{1+\gamma} + (1 - \alpha) \left( \frac{z - z_a(1 + t)}{w_b} \right)^{1+\gamma}.$$

Now, if we define  $\hat{w}_a \equiv w_a(1 + t)^{\frac{\gamma}{1+\gamma}}$ , then it is not hard to check that

$$\hat{k}_a(\boldsymbol{\iota}) = \frac{\hat{w}_a(1 - \alpha)^{1/\gamma}}{\hat{w}_a(1 - \alpha)^{1/\gamma} + w_b\alpha^{1/\gamma}}, \quad \hat{k}_b(\boldsymbol{\iota}) = 1 - \hat{k}_a(\boldsymbol{\iota}),$$

$$\hat{\theta}(\boldsymbol{\iota}) := \frac{\hat{w}(\boldsymbol{\iota})^{-1-\gamma}}{[\alpha^{1/\sigma} + (1 - \alpha)^{1/\sigma}]^\sigma},$$

for

$$\hat{w}(\boldsymbol{\iota}) := \left[ \alpha^{\frac{-1}{\gamma}} \hat{w}_a^{\frac{1+\gamma}{\gamma}} + w_b^{\frac{1+\gamma}{\gamma}} (1 - \alpha)^{\frac{-1}{\gamma}} \right]^{\frac{\gamma}{1+\gamma}}.$$

For every  $\theta$ , define  $\mathcal{I}_t(\theta) \equiv \left\{ \boldsymbol{\iota} \in \Lambda \mid \hat{\theta}(\boldsymbol{\iota}) = \theta \right\}$ , then, let

$$\bar{k}_a(\theta) = \mathbb{E} \left[ \hat{k}_a(\boldsymbol{\iota}) \mid \boldsymbol{\iota} \in \mathcal{I}_t(\theta) \right], \quad \bar{k}_b(\theta) = \mathbb{E} \left[ \hat{k}_b(\boldsymbol{\iota}) \mid \boldsymbol{\iota} \in \mathcal{I}_t(\theta) \right].$$

Letting  $\hat{\psi}$  denote the distribution of  $\theta$  induced by the identity  $\theta \equiv \hat{\theta}(\boldsymbol{\iota})$ , given the distribution of  $\boldsymbol{\iota}$ , we can write the cost of this policy as<sup>26</sup>

$$t \int_{\underline{\theta}}^{\bar{\theta}} \bar{k}_a(\theta) z(\theta) \hat{\psi}(\theta) d\theta.$$

When compared to the planner's program in page 16, the only differences are the distribution of types which is now  $\hat{\psi}$  and the budget constraint which now includes the cost of financing the subsidy to spouse  $a$ .

be subjected to an individual linear income tax with constant (positive or negative) rate  $t \in \mathbb{R}$ . Note that this tax is raised on spouse  $a$ 's gross income leading to a new value,  $z = z_a(1 - t) + z_b$ , for household taxable income. This type of taxation adds gender-based incentives to the tax system since the marginal tax rate faced by each spouse is not the same, as opposed to our baseline case. Note that

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<sup>26</sup>Bear in mind that the support of  $\hat{\psi}$  may differ from that of  $\psi$ .

the marginal tax rate on the income produced by spouse  $a$  is not  $T'(z) + t$ , but  $T'(z)(1 - t) + t$ .

Moreover, this tax shifts effort toward spouse  $b$ . Indeed, it is straightforward to check that

$$\frac{z_a}{z_b} = \left[ \frac{(1 - \alpha)(1 - t)}{\alpha} \left( \frac{w_a}{w_b} \right)^{1+\gamma} \right]^{\frac{1}{\gamma}}. \quad (18)$$

For small tax  $t$ , spouse  $b$  is "relatively more productive".

To understand the impact of  $t$  on household income, it is worth noting that household preferences can once again be represented by a utility function of the form

$$V(x, z, \theta_a, \theta_b, \alpha, t) = \kappa(\alpha) + \ln x - \theta(w_a, w_b, \alpha, t) \frac{z^{1+\gamma}}{1 + \gamma},$$

where  $\kappa(\alpha) \equiv \alpha \ln(\alpha) + (1 - \alpha) \ln(1 - \alpha)$  is fully determined by  $\alpha$  and

$$\begin{aligned} \omega(\theta_a, \theta_b, \alpha, t) \equiv & \alpha \theta_a \left[ \frac{[(1 - \alpha)\theta_b(1 - t)]^{\frac{1}{\gamma}}}{[(1 - \alpha)\theta_b(1 - t)]^{\frac{1}{\gamma}} + [\alpha\theta_a]^{\frac{1}{\gamma}}} \right]^{1+\gamma} + \\ & (1 - \alpha)\theta_b \left[ \frac{[\alpha\theta_a]^{\frac{1}{\gamma}}}{[(1 - \alpha)\theta_b(1 - t)]^{\frac{1}{\gamma}} + [\alpha\theta_a]^{\frac{1}{\gamma}}} \right]^{1+\gamma} \end{aligned}$$

is a family type unitary type and a function of spouses' types, the Pareto weight, and the tax rate.

The tax  $t$  on  $\omega$  affects household productivity by increasing the relative importance of spouse  $b$ 's productivity. Hence,  $\omega$  may increase with  $t$  or decrease with  $t$  depending on  $(1 - \alpha)\theta_b > \alpha\theta_a$  and on  $\gamma$ .

We can further simplify the expressions above if we define

$$\zeta_a(t) = \frac{[(1 - \alpha)\theta_b(1 - t)]^{\frac{1}{\gamma}}}{[(1 - \alpha)\theta_b(1 - t)]^{\frac{1}{\gamma}} + [\alpha\theta_a]^{\frac{1}{\gamma}}} \quad \text{and} \quad \zeta_b(t) = \frac{[\alpha\theta_a]^{\frac{1}{\gamma}}}{[(1 - \alpha)\theta_b(1 - t)]^{\frac{1}{\gamma}} + [\alpha\theta_a]^{\frac{1}{\gamma}}}.$$

In fact, we get  $z_a = \zeta_a z$ ,  $z_b = \zeta_b z$  and  $\omega(t) = \alpha\theta_a\zeta_a(t)^{1+\gamma} + [1 - \alpha]\theta_b\zeta_b(t)^{1+\gamma}$ . Note that the marginal tax rate faced by each household in the definition of the program is also changed by  $t$ . Indeed, we can use (18) to write the household budget constraint as

$$x = z [1 - t\zeta_a(t)] - T(z [1 - t\zeta_a(t)]).$$

This shows that, although household productivity  $\omega$  may be increased by  $t$ , the marginal tax rate faced by the household is always increased by  $t$ . It is not hard to

show, by simple envelope arguments, that the impact on the utility *as perceived by the household* is unambiguously negative. In the absence of dissonance, this negative impact is the one that must be compared with the positive revenue impact to assess the value of a small tax reform.

When dissonance is taken into account, however, the utility *as perceived by the planner* need not decrease. We have seen that  $t$  acts as a subsidy on spouse  $a$ 's leisure. Moreover,  $t$  has no impact on how total consumption is shared by spouses. Hence, a positive  $t$  produces a redistribution of utility from spouse  $b$  to spouse  $a$ , which is valued by the planner when  $\alpha < 1/2$ . This welfare gain must be, in general, compared with the overall loss associated with higher taxes.

Alternatively, note that a small tax  $t$  on  $z_a$  will increase tax revenues. To keep the balanced budget, one could simply reduce  $T(\cdot)$  by an amount  $H$  such that

$$tz\zeta_a(t) + T(z(1 - tz\zeta_a(t))) - H(z(1 - tz\zeta_a(t))) = T(z).$$

Recall that since  $\alpha$  is the same for all couples, a  $(\theta_a, \theta_b, \alpha)$  couple may be equally represented by  $(\omega, \omega_a)$ . Using this representation, the introduction of a small  $t$  must be accompanied by a decrease in marginal tax rates captured by

$$h(\omega, \omega_a) = \hat{\zeta}_a(\omega, \omega_a)z[1 - T'(z)],$$

where  $\hat{\zeta}_a(\omega, \omega_a) = \zeta_a(0)$  for a type  $(\omega, \omega_a)$  couple.

Two things about this expression are worth noting. First, the reduction in the marginal tax rate would have to differ for two couples with the same earnings if the share earned by  $a$  differs. Second, because  $z$  is a function of  $\omega$  only, all the dependency on  $\omega_a$  is through  $\hat{\zeta}_a(\omega, \omega_a)$ . This allows us to write the feasible reform in which

$$H(z(\omega)) = z(\omega)[1 - T'(z(\omega))]\mathbb{E}_{\omega_a} \left[ \hat{\zeta}_a(\omega, \omega_a) | \omega \right].$$

Such reform distributes income from couples with a high participation of  $z_a$  on total income (high  $\hat{\zeta}_a(\omega, \omega_a)$ ) toward couples with low participation. Since we are holding  $\alpha$  fixed across households, this redistribution is toward couples with very asymmetric productivities between spouses. With log utility for consumption, the planner weights money in the hands of couples only by  $z$  and not by how it is distributed across spouses. Hence, the merits of this reform only depend on whether the redistribution of utility from  $b$  to  $a$  justifies the inefficiency generated by the misallocation of effort across spouses. If the planner's metric puts more weight on spouse  $a$  than couples do, then, for a small enough  $t$ , the answer is yes. In Appendix H we show that the presence of household production need not overturn the results and most likely makes it more severe.

## 5.2 A parametric tax function

Assume that the initial joint tax schedule is of the form used by [Musgrave \(1959\)](#), [Bénabou \(2000\)](#), and [Heathcote et al. \(2017\)](#) ( i.e.,  $T(z) = z - \chi z^{1-\kappa}$ ). Then, assume that a small tax  $t$  is introduced in the income earned by spouse  $a$ . The tax base is now  $\tilde{z} = z [1 - t\zeta_a(t)]$ . The household budget constraint becomes

$$x = z [1 - t\zeta_a(t)] - \{z [1 - t\zeta_a(t)] - \chi [z [1 - t\zeta_a(t)]]^{1-\kappa}\} = \chi [1 - t\zeta_a(t)]^{1-\kappa} z^{1-\kappa}.$$

The impact on the household constraint is to reduce  $\chi$  to  $\chi [1 - t\zeta_a(t)]^{1-\kappa}$ . Given the preferences we are using (log in consumption), this has no impact on household labor supply. Indeed,

$$z = \left[ \frac{1 - \kappa}{\omega(\theta_a, \theta_b, \alpha, t)} \right]^{\frac{1}{1+\gamma}}$$

is the same expression we would obtain without  $t$ , if  $\omega$  were held fixed. So, the effect of  $t$  on labor supply is completely captured by how taxes change  $\omega$ .

[ FIGURE 6 ABOUT HERE ]

Figure 6, left panel, displays the impact on each spouse's utility and on  $V$ , the planner's assessment of household utility, of introducing a small tax  $\tau_a = 5\%$  on spouse  $a$ 's earnings. It is apparent that spouse  $a$  always benefits. Since  $\chi$  is adjusted to hold total government revenues fixed, there is also a redistribution from households in which spouse  $a$ 's earnings are a higher proportion of household income to households in which it is a lower proportion. The right panel makes it clear that the "intuitive" policy of subsidizing  $a$  reduces her welfare.

### Policy-dependent threat points

[Immervoll et al. \(2011\)](#), who make the same independence assumption regarding the Pareto weights, justify it by assuming that weights depend on the utilities attained by singles, which, they claim, are unaffected by the taxation of couples. Although this argument has merit, one must also recognize that, under a collective approach, changes in a household budget set may have an indirect impact on choices through changes in the household objective. In our model, this amounts to  $\alpha$  possibly being affected by policy.

We show how this collective feature may overturn the counterintuitive finding from the previous policy evaluations and restore the more commonsense view that taxing wives hurts them.

Let  $z = z_a(1 - \tau) + z_b$ . Note that the gross income produced by such household is  $\hat{z} = z_a + z_b = z + \tau z_a$ . Since  $y = z_a(1 - \tau) + z_b - T(z_a(1 - \tau) + z_b) = z - T(z)$ , we shall focus on this variable for the purpose of understanding household choices. To simplify the terminology we shall refer to  $z = z_a(1 - \tau) + z_b$  as earnings and  $\hat{z} = z_a + z_b$  as gross income.

Spouses generate earnings efficiently by solving

$$\mathcal{V}_\tau(z, \alpha, w_a, w_b) = \min_{z_a} \left\{ \alpha \left( \frac{z_a}{w_a} \right)^{1+\gamma} + (1 - \alpha) \left( \frac{z - z_a(1 - \tau)}{w_b} \right)^{1+\gamma} \right\},$$

which has, as a first-order condition,

$$\frac{\alpha}{(1 - \alpha)(1 - \tau)} \frac{w_b^{1+\gamma}}{w_a^{1+\gamma}} z_a^\gamma = [z - z_a(1 - \tau)]^\gamma.$$

Now, the crucial assumption we make in the spirit of the collective approach is that<sup>27</sup>

$$\alpha = \frac{w_a(1 - \tau)}{w_a(1 - \tau) + w_b}.$$

In this case, we find that the wife's earnings as a share of earnings,  $z$ , are

$$z_a = \frac{w_a}{w_b + w_a(1 - \tau)} z,$$

which we could also have written as a function of the household's gross income,  $\hat{z}$ , as  $z_a = w_a \hat{z} / (w_b + w_a)$ .

A little bit of algebra allows us to write the household aggregated disutility of effort,  $\alpha n_a^{1+\gamma} / (1 + \gamma) + (1 - \alpha) n_b^{1+\gamma} / (1 + \gamma)$ , as

$$\hat{\mathcal{V}}_\tau(w_a, w_b) = \frac{1}{1 + \gamma} \left( \frac{z}{w_b + w_a(1 - \tau)} \right)^{1+\gamma}.$$

Now, letting  $\omega_\tau(w_a, w_b) = (w_a(1 - \tau) + w_b)^{-1-\gamma}$ , then the household optimal earnings decision problem can be written more simply as

$$\max_z \left\{ \ln(z - T(z)) - \omega_\tau(w_a, w_b) \frac{z^{1+\gamma}}{1 + \gamma} \right\}.$$

Introducing a gender-dependent tax,  $\tau$ , involves three consequences for the gender being taxed. First, as before, the reward from this spouse's effort is less rewarded. This is a force toward increasing her leisure. However, the "collective"

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<sup>27</sup>The collective approach allows the share of household income that is received by each spouse to affect the allocation. In empirical works, exogenous variables – usually unearned income – are used. Here, we use *net* productivity to affect weights.

nature of couples means that as the spouse's ability to contribute to the couple reduces, so does her entitlement to the household surplus. Under the specific assumption we have made about  $\alpha$  as a function of  $(w_a(1 - \tau), w_b)$ , the amount of effort (hence, leisure) per unit of gross income  $\hat{z} = z_a + z_b$  is not changed by taxes. Second, the tax is, for all purposes, making the household less productive from a private perspective since a unit of gross income  $\hat{z}$  is converted to  $z = \hat{z} - \tau z_a < \hat{z}$  units of taxable income. Finally, a lower  $\alpha$  implies that spouse  $a$  gets a smaller share of  $z - T(z)$ .

Hence, in contrast with the case most frequently explored in the literature, if one takes the collective nature of households seriously, the "intuitive" result reappears.

### 5.3 Children

Although we have focused on the case in which society quarrels about the weights that spouses assign each other, a potentially significant issue relates to the treatment of children. As emphasized by [Farhi and Werning \(2007\)](#), a utilitarian objective must consider the welfare of future generations not only through their impact on the parents' welfare but also directly as separate beings.

Many income support programs have as the primary goal to improving the situation of children. In particular, policies aimed at empowering women with the explicit intent of helping children's development are found in programs like *Bolsa Família* in Brazil and *Oportunidades* in Mexico. The idea behind those programs is that women have a higher degree of altruism towards their children, and direct transfers to mothers reflect in higher expenditure on children. Here we will briefly illustrate how incorporating children and different degrees of altruism of mothers and fathers toward their children can also generate dissonance.

Assume for the sake of the illustration that spouses have caring preferences toward one another and their children. In fact, assume that spouse  $i$ 's utility is of the form

$$U_i = U_i + \delta_i U_j + \delta_i^c U^c, \quad i, j = a, b.,$$

where  $U_i$  is the flow utility that  $i$  obtains from his/her consumption bundle,  $U_j$ , is the total utility of  $i$ 's spouse and  $U^c$  is the flow utility that their child attains from his/her consumption bundle.  $0 \leq \delta_i < 1$  is the degree of  $i$ 's altruism towards his/her spouse, and  $0 < \delta_i^c < 1$  towards the child. We can, in this case, re-express spouses' utilities as

$$U_a = \frac{U_a + \delta_a U_b + \hat{\delta}_a^c U^c}{1 - \delta_a \delta_b}, \quad \text{and,} \quad U_b = \frac{U_b + \delta_b U_a + \hat{\delta}_b^c U^c}{1 - \delta_a \delta_b},$$



where  $\hat{\delta}_i^c = \delta_i^c + \delta_i \delta_c^j$ .

Now we can incorporate children by both takings seriously the possible different degrees of altruism of mothers and fathers toward their children and the direct value of children's welfare in planner's objective, as highlighted by Farhi and Werning (2007). In that case, spouses choose according to  $V = \alpha U_a + (1 - \alpha) U_b$ , whereas the planner assesses welfare according to  $W = [U_a + U_b + U^c]/3$ .

If we focus on egotistic (towards each other) preferences, we can take children into account by writing

$$U_i(x_i, x^c, l_i) = \ln x_i + \delta_i \ln x^c - \frac{l_i^{1+\gamma}}{1+\gamma},$$

where  $x^c$  is the child's consumption and  $0 < \delta_i < 1$  is the degree of altruism of spouse  $i$  toward the child.

In this case, total expenditure in the child's consumption as a fraction of household disposable income  $x$  is given by  $\delta/(1+\delta)$  for  $\delta = \alpha\delta_a + (1-\alpha)\delta_b$ . The inclusion of preferences for children affects the preferences for  $(x, z)$  bundles by a household by increasing the value of consumption by a factor  $\delta$  – the marginal rate of substitution is now  $z^\gamma x \omega / (1 + \delta)$ .

In contrast, we assume that the planner assigns weights  $1/3$  for each spouse and the child. This leads to a valuation of the form  $z^\gamma x^\sigma \tilde{\omega} / (1.5 + \tilde{\delta})$ , where  $\tilde{\omega} = \omega(1/2, \theta_a, \theta_b)$ , and  $\tilde{\delta} = (\delta_a + \delta_b)/2$ .<sup>28</sup> Beyond the difference between  $\omega = \hat{\omega}$ , already explored, two novel sources of divergence between planner and spouses' preferences arise. First, is the fact that the relevant impact of children's welfare on spouses' utilities is evaluated by the planner under the assumption of equal weights, while spouses use  $\alpha$ . Second, which exists even if the planner assigns weights  $\alpha$  and  $1 - \alpha$ , is the direct impact of child's utility, which is captured by the value 1.5 instead of 1 in the marginal rate of substitution.

**Public Goods** Note that children enter household preferences as public goods. Children are not, of course, the only public goods that spouses consume. However, they are unique from a normative perspective since society values children above and beyond their impact on spouses' well-being.

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<sup>28</sup>The planner would like the family to solve

$$\max \frac{1}{3} U_a(x_a, x^c, l_a) + \frac{1}{3} U_b(x_b, x^c, l_b) + \frac{1}{3} \ln x^c.$$

## 6 Conclusion

In this paper, we obtain optimal tax schedules for multiperson households under the restriction that taxes entail income-splitting. Intrahousehold inequality, a policy goal often ignored in the study of redistribution policies following the Mirrlees tradition, takes center stage.

The use of a common specification for preferences allows us to handle the multidimensional screening problem that plagues optimal household taxation and to actually compute the optimal tax schedule. Our results highlight the quantitative impact of dissonance, the misalignment between the household's and planner's objectives for household income taxes: an instrument that can only indirectly affect intrahousehold inequality. Optimal marginal tax rates need no longer be non-negative which is in contrast with a well-known property derived for single-person households in the Mirrlees tradition.

Departing from a joint schedule, we explore the consequences of introducing some form of differentiation in the marginal tax rates faced by the two spouses. We show that it is typically optimal to introduce a *small* tax on the spouse that society tries to promote, since income taxes subsidize spouse's leisure. This crucially depends on the assumption that power is unaffected by choices or the tax system (or both). It is clear that if earnings – a policy-dependent variable – instead of innate productivity – a policy-independent variable – determine power, these results may be reversed.

Finally, to take into account the potential heterogeneity of preferences across agents, we derive optimal tax formulae preferences using tax perturbation methods. Optimal tax formulae have an ABC+D expression that extends the ABC formula made popular by [Diamond \(1998\)](#) by introducing an additive D term: a Pigouvian correction for dissonance. This Pigouvian term, which adjusts the marginal tax rate, has an analytical expression for the special case of logarithmic preferences for consumption. We find that, for small values of dissonance, it decreases from a value of zero as more power is given to the most productive spouse. This suggests that if bargaining power within the household is positively correlated with relative productivity, then marginal tax rates on couples should be lower than their counterpart for singles.

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## A Lemmatta

### A.1 Collective decision

#### The consumption allocation problem within the household

**Lemma 1.** *The efficient solution to the household consumption allocation problem is  $x_a = sx$  and  $x_b = (1 - s)x$ , for  $s = \alpha^{1/\sigma}(\alpha^{1/\sigma} + (1 - \alpha)^{1/\sigma})^{-1}$ .*

*Proof.* The disposable income  $x = z - T(z)$  is allocated to the consumption of private goods of the two spouses according to

$$\max_{(x_a, x_b) \in \mathbb{R}_{++}^2} \alpha \frac{x_a^{1-\sigma}}{1-\sigma} + (1-\alpha) \frac{x_b^{1-\sigma}}{1-\sigma}$$

subject to

$$x_a + x_b = x.$$

Defining  $s = x_a/x$  at the value for  $x_a$  that satisfies the first order condition for this program yields gives us the result.  $\square$

**Lemma 2.** *Each spouse's earned income  $z_i$ , for  $i \in \{a, b\}$ , can be expressed in terms of household income  $z$ , productivity gap  $\theta_a/\theta_b$  and the Pareto coefficient  $\alpha$  as*

$$\frac{z_a}{z} = \frac{[(1-\alpha)\theta_b]^{\frac{1}{\gamma}}}{[(1-\alpha)\theta_b]^{\frac{1}{\gamma}} + [\alpha\theta_a]^{\frac{1}{\gamma}}} \quad \text{and} \quad \frac{z_b}{z} = \frac{[\alpha\theta_a]^{\frac{1}{\gamma}}}{[(1-\alpha)\theta_b]^{\frac{1}{\gamma}} + [\alpha\theta_a]^{\frac{1}{\gamma}}}.$$

*Proof.* Given our assumptions of efficient allocation within the household, the division of labor supply is made in order to minimize the disutility costs of getting a given income level  $z \in \mathbb{R}_+$ . Therefore, the collective choice of labor supply is the solution to

$$\min_{z_a, z_b} \alpha \left( \theta_a \frac{z_a^{1+\gamma}}{1+\gamma} \right) + (1-\alpha) \left( \theta_b \frac{z_b^{1+\gamma}}{1+\gamma} \right)$$

subject to

$$z_a + z_b = z.$$

The first-order condition for this problem is

$$\frac{\alpha}{1-\alpha} \frac{\theta_a}{\theta_b} = \left( \frac{z_b}{z_a} \right)^\gamma. \quad (\text{A.1})$$

Using  $z = z_a + z_b$ , we have

$$z_a = z \left[ \left( \frac{\alpha}{1-\alpha} \frac{\theta_a}{\theta_b} \right)^{\frac{1}{\gamma}} + 1 \right]^{-1} \quad \text{and} \quad z_b = z \left[ \left( \frac{1-\alpha}{\alpha} \frac{\theta_b}{\theta_a} \right)^{\frac{1}{\gamma}} + 1 \right]^{-1}. \quad (\text{A.2})$$

This proves the result.  $\square$

**Lemma 3.** Suppose  $\sigma = 1$ , the household welfare function for a type  $(\theta_a, \theta_b, \alpha)$  family can be expressed as a function of total earned income  $z$  and disposable income  $x$  as

$$V(x, z, \theta_a, \theta_b, \alpha) = \kappa(\alpha) + \ln x - \omega(\theta_a, \theta_b, \alpha) \frac{z^{1+\gamma}}{1+\gamma},$$

where  $\kappa(\alpha) \equiv \alpha \ln(\alpha) + (1 - \alpha) \ln(1 - \alpha)$  and

$$\begin{aligned} \omega(\theta_a, \theta_b, \alpha) \equiv & \alpha \theta_a \left( \frac{[(1 - \alpha)\theta_b]^{\frac{1}{\gamma}}}{[(1 - \alpha)\theta_b]^{\frac{1}{\gamma}} + [\alpha\theta_a]^{\frac{1}{\gamma}}} \right)^{1+\gamma} \\ & + (1 - \alpha)\theta_b \left( \frac{[\alpha\theta_a]^{\frac{1}{\gamma}}}{[(1 - \alpha)\theta_b]^{\frac{1}{\gamma}} + [\alpha\theta_a]^{\frac{1}{\gamma}}} \right)^{1+\gamma}. \end{aligned}$$

*Proof.* Using Lemmas 1 and 2 we can restate the family utility in (2) as

$$\begin{aligned} V(x, z, \theta_a, \theta_b, \alpha) = & \alpha \ln \alpha + (1 - \alpha) \ln(1 - \alpha) + \ln x \\ & - \left\{ \alpha \theta_a \left[ 1 + \left( \frac{\alpha}{1 - \alpha} \frac{\theta_a}{\theta_b} \right)^{\frac{1}{\gamma}} \right]^{-(1+\gamma)} + (1 - \alpha) \theta_b \left[ 1 + \left( \frac{1 - \alpha}{\alpha} \frac{\theta_b}{\theta_a} \right)^{\frac{1}{\gamma}} \right]^{-(1+\gamma)} \right\} \frac{z^{1+\gamma}}{1 + \gamma} \end{aligned}$$

defining  $\omega(\theta_a, \theta_b, \alpha)$  and  $\kappa(\alpha)$  as in the statement of the lemma and we have the result proved.  $\square$

**Lemma 4.** The indirect utility of spouses in a family with total income  $z$ , disposable income  $x$ , types  $\theta_a, \theta_b$  and Pareto weight  $\alpha$  for the wife can be written as

$$\begin{aligned} V_a(x, z, \theta_a, \theta_b, \alpha) = & \ln \alpha - \kappa(\alpha) + V(x, z, \theta_a, \theta_b, \alpha) + [\omega - \omega_a] \frac{z^{1+\gamma}}{1 + \gamma} \quad \text{and} \\ V_b(x, z, \theta_a, \theta_b, \alpha) = & \ln(1 - \alpha) - \kappa(\alpha) + V(x, z, \theta_a, \theta_b, \alpha) + [\omega - \omega_b] \frac{z^{1+\gamma}}{1 + \gamma}, \end{aligned}$$

where  $V(x, z, \theta_a, \theta_b, \alpha)$ ,  $\kappa(\alpha)$  and  $\omega(\theta_a, \theta_b, \alpha)$  are defined in (??) and (8), respectively,

$$\begin{aligned} \omega_a(\theta_a, \theta_b, \alpha) \equiv & \theta_a \left( \frac{[(1 - \alpha)\theta_b]^{\frac{1}{\gamma}}}{[(1 - \alpha)\theta_b]^{\frac{1}{\gamma}} + [\alpha\theta_a]^{\frac{1}{\gamma}}} \right)^{1+\gamma} \quad \text{and} \\ \omega_b(\theta_a, \theta_b, \alpha) \equiv & \theta_b \left( \frac{[\alpha\theta_a]^{\frac{1}{\gamma}}}{[(1 - \alpha)\theta_b]^{\frac{1}{\gamma}} + [\alpha\theta_a]^{\frac{1}{\gamma}}} \right)^{1+\gamma}. \end{aligned} \quad (\text{A.3})$$

*Proof.* Let  $V_a(x, z, \theta_a, \theta_b, \alpha)$  and  $V_b(x, z, \theta_a, \theta_b, \alpha)$  be indirect utility derived by spouses;  $a$  and  $b$  in a family type  $(\theta_a, \theta_b, \alpha)$  from the aggregate bundle  $(x, z)$ . Using Lemmas 1

and 2, we have

$$V_a(x, z, \theta_a, \theta_b, \alpha) = \ln \alpha x - \omega_a(\theta_a, \theta_b, \alpha) \frac{z^{1+\gamma}}{1+\gamma} \quad \text{and}$$

$$V_b(x, z, \theta_a, \theta_b, \alpha) = \ln(1-\alpha)x - \omega_b(\theta_a, \theta_b, \alpha) \frac{z^{1+\gamma}}{1+\gamma},$$

where

$$\omega_a(\theta_a, \theta_b, \alpha) \equiv \theta_a \left( \frac{[(1-\alpha)\theta_b]^{\frac{1}{\gamma}}}{[(1-\alpha)\theta_b]^{\frac{1}{\gamma}} + [\alpha\theta_a]^{\frac{1}{\gamma}}} \right)^{1+\gamma} \quad \text{and}$$

$$\omega_b(\theta_a, \theta_b, \alpha) \equiv \theta_b \left( \frac{[\alpha\theta_a]^{\frac{1}{\gamma}}}{[(1-\alpha)\theta_b]^{\frac{1}{\gamma}} + [\alpha\theta_a]^{\frac{1}{\gamma}}} \right)^{1+\gamma}.$$

It is straightforward to see from these definitions that

$$\omega(\theta_a, \theta_b, \alpha) = \alpha\omega_a(\theta_a, \theta_b, \alpha) + (1-\alpha)\omega_b(\theta_a, \theta_b, \alpha).$$

Using the definition of  $V(x, z, \theta_a, \theta_b)$  from Lemma 3 and neglecting the dependence for shortness

$$\begin{aligned} V &= \kappa(\alpha) + \ln x - \omega \frac{z^{1+\gamma}}{1+\gamma} = \kappa(\alpha) + \ln \left( \alpha x \frac{1}{\alpha} \right) - [\alpha\omega_a + (1-\alpha)\omega_b] \frac{z^{1+\gamma}}{1+\gamma} \\ &= \kappa(\alpha) - \ln(\alpha) - (1-\alpha)[\omega_b - \omega_a] \frac{z^{1+\gamma}}{1+\gamma} + \left[ \ln(\alpha x) - \omega_a \frac{z^{1+\gamma}}{1+\gamma} \right] \\ &= \kappa(\alpha) - \ln(\alpha) - (1-\alpha)[\omega_b - \omega_a] \frac{z^{1+\gamma}}{1+\gamma} + V_a. \end{aligned}$$

Notice that

$$(1-\alpha)[\omega_b - \omega_a] = [(1-\alpha)\omega_b + \alpha\omega_a] - \omega_a = \omega - \omega_a.$$

Rearranging terms we have

$$V_a = \ln(\alpha) - \kappa(\alpha) + V + [\omega - \omega_a] \frac{z^{1+\gamma}}{1+\gamma}.$$

Analogously,

$$V_b = \ln(1-\alpha) - \kappa(\alpha) + V + [\omega - \omega_b] \frac{z^{1+\gamma}}{1+\gamma},$$

where all functions are evaluated at  $(\theta_a, \theta_b, \alpha)$ . □



**Lemma 5.** *Suppose that the social welfare function follows the household's utility (as in (2)). In an interval where  $z(\omega)$  is strictly increasing, the marginal tax rate for the optimal nonlinear income joint taxation will follow exactly as in Mirrlees' model:*

$$T'(z(\omega)) = \frac{z(\omega)^\gamma}{\psi(\omega)} \int_{\omega}^{\bar{\omega}} \left[ x(s) - \frac{1}{\lambda} \right] \psi(s) ds, \quad (\text{A.4})$$

where  $\lambda \in \mathbb{R}_+$  is the Lagrange multiplier associated with the budget constraint.

*Proof of Lemma 5.* By the revelation principle, the planner's problem based solely on the family's utility can be restated in the space of direct mechanism as<sup>29</sup>

$$\max_{(x,z):\Omega \rightarrow \mathbb{R}_{++} \times \mathbb{R}_+} \int_{\Omega} \left[ \ln x(\omega) - \omega \frac{z(\omega)^{1+\gamma}}{1+\gamma} \right] \psi(\omega) d\omega,$$

subject to incentive compatibility constraints: for every  $\omega \in \Omega$

$$\omega \in \arg \max_{\tilde{\omega} \in \Omega} \left[ \ln x(\tilde{\omega}) - \tilde{\omega} \frac{z(\tilde{\omega})^{1+\gamma}}{1+\gamma} \right],$$

and a budget constraint:

$$\int_{\Omega} [z(\omega) - x(\omega)] \psi(\omega) d\omega \geq 0.$$

Let  $v(\omega)$  denote the utility assignment of a family with type  $\omega$  in the mechanism. Taking  $v(\omega)$  and  $z(\omega)$  as the choice variables,  $x(\omega)$  is implicitly defined by

$$v(\omega) = \ln x(v(\omega), z(\omega), \omega) - \omega \frac{z(\omega)^{1+\gamma}}{1+\gamma}.$$

First, the family's utility satisfies the *Spence-Mirrlees condition* with respect to the parameter  $\omega$  since

$$\left. \frac{dx}{dz} \right|_{v(\omega)=v} = \frac{h'(z)}{u'(x)} = \omega x z^\gamma,$$

which means that the slope of the indifference curve is increasing in the parameter  $\omega$ . Hence, incentive compatibility can be fully characterized by the local constraints. The planner solves the following program:

$$\max_{(v,z):\Omega \rightarrow \mathbb{R} \times \mathbb{R}_+} \int_{\Omega} v(\omega) \psi(\omega) d\omega$$

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<sup>29</sup>Since the term  $\kappa(\alpha)$  is not affected by the allocation we can ignore it.

subject to the local first-order incentive constraint:

$$\dot{v}(\omega) = V_\omega(x, z, \omega) = \frac{z(\omega)^{1+\gamma}}{1+\gamma}; \quad (\text{A.5})$$

the monotonicity constraint:  $z(\omega)$  is increasing; and the feasibility constraint:

$$\int_{\Omega} [z(\omega) - x(v(\omega), z(\omega), \omega)] \psi(\omega) d\omega = 0.$$

By the implicit function theorem,

$$x_v = \frac{1}{u'(x(v(\omega), z(\omega), \omega))} = x(v(\omega), z(\omega), \omega), \quad (\text{A.6})$$

$$x_z = \frac{h'(z(\omega))}{u'(x(v(\omega), z(\omega), \omega))} = \omega z(\omega)^\gamma x(v(\omega), z(\omega), \omega). \quad (\text{A.7})$$

Under this approach, the income  $z(\omega)$  is the control variable and utility of the family  $v(\omega)$  is the state variable. Hence, the Hamiltonian associated to this problem is

$$\begin{aligned} \mathcal{H}(v(\omega), z(\omega), \mu(\omega), \lambda, \omega) = & v(\omega) \psi(\omega) + \\ & \lambda [z(\omega) - x(v(\omega), z(\omega), \omega)] \psi(\omega) - \mu(\omega) \frac{z(\omega)^{1+\gamma}}{1+\gamma}, \end{aligned}$$

where  $\lambda \in \mathbb{R}$  is the Lagrange multiplier associated to the resources constraint and  $\mu(\omega)$  is the co-state variable. At the optimum, the control variable  $z(\omega)$  maximizes the Hamiltonian function:

$$\lambda \psi(\omega) [1 - x_z] = \mu(\omega) z(\omega)^\gamma. \quad (\text{A.8})$$

The first-order condition of the Hamiltonian is

$$\dot{\mu}(\omega) = -\frac{\partial}{\partial v} \mathcal{H} = \psi(\omega) [\lambda x_v - 1]$$

and the transversality condition is  $\mu(\underline{\omega}) = \mu(\bar{\omega}) = 0$ . Integrating on both sides from  $\omega$  to  $\bar{\omega}$ , we have

$$\mu(\bar{\omega}) - \mu(\omega) = \int_{\omega}^{\bar{\omega}} \dot{\mu}(s) ds = \int_{\omega}^{\bar{\omega}} \psi(s) [\lambda x_v(v(s), z(s), s) - 1] ds.$$

Using the transversality condition we have

$$\mu(\omega) = \lambda \int_{\omega}^{\bar{\omega}} \psi(s) [\lambda^{-1} - x_v(v(s), z(s), s)] ds. \quad (\text{A.9})$$

Note that from the transversality condition  $\mu(\underline{\omega}) = 0$  we can pin down the social value of an extra unit of income:

$$0 = \mu(\underline{\omega}) = \lambda \int_{\underline{\omega}}^{\bar{\omega}} \psi(s) [\lambda^{-1} - x_v(v(s), z(s), s)] ds.$$

Therefore,

$$\lambda = \frac{1}{\int_{\underline{\omega}}^{\bar{\omega}} x_v(v(s), z(s), s) \psi(s) ds} = \frac{1}{\int_{\underline{\omega}}^{\bar{\omega}} x(v(s), z(s), s) \psi(s) ds}$$

or

$$\lambda = \mathbb{E}^{\psi}[x]^{-1}.$$

Substituting  $\mu(\omega)$  defined in (A.9) into equation (A.8) we have

$$[1 - x_z(v(\omega), z(\omega), \omega)] = \frac{z(\omega)^{\gamma}}{\psi(\omega)} \int_{\omega}^{\bar{\omega}} [\lambda^{-1} - x_v(v(s), z(s), s)] \psi(s) ds.$$

This gives us an expression for the optimal joint income tax rate that follows exactly the one of the standard Mirrlees' model:

$$T'(z(\omega)) = \frac{z(\omega)^{\gamma}}{\psi(\omega)} \int_{\omega}^{\bar{\omega}} [x_v(v(s), z(s), s) - \lambda^{-1}] \psi(s) ds, \quad (\text{A.10})$$

whenever this differential equation gives an increasing function  $z(\omega)$ . Using  $\lambda = \mathbb{E}^{\psi}[x]^{-1}$  and  $x_v = x(v(\omega), z(\omega), \omega)$ , we can rewrite, with some abuse of notation ( $x_v(v(s), z(s), s) = x(s)$ ), the formula for the optimal marginal taxation as

$$T'(z(\omega)) = \frac{z(\omega)^{\gamma}}{\psi(\omega)} \int_{\omega}^{\bar{\omega}} [x(s) - \mathbb{E}^{\psi}[x]] \psi(s) ds. \quad (\text{A.11})$$

□

**Lemma 6.** *Suppose that there is a introduction of a small tax  $t$  in the spirit of Subsection 5.1. The household welfare function for a type  $(\theta_a, \theta_b, \alpha)$  family can once again be expressed as a function of total earned income,  $z$  and disposable income  $x$  as*

$$V(x, z, \theta_a, \theta_b, \alpha, t) = \kappa(\alpha) + \ln x - \omega(\theta_a, \theta_b, \alpha, t) \frac{z^{1+\gamma}}{1+\gamma},$$

where  $\kappa(\alpha) \equiv \alpha \ln(\alpha) + (1 - \alpha) \ln(1 - \alpha)$  is fully determined by  $\alpha$ , hence exogenous, and

$$\omega(\theta_a, \theta_b, \alpha, t) \equiv \alpha \theta_a \left( \frac{[(1 - \alpha)\theta_b(1 - t)]^{\frac{1}{\gamma}}}{[(1 - \alpha)\theta_b(1 - t)]^{\frac{1}{\gamma}} + [\alpha\theta_a]^{\frac{1}{\gamma}}} \right)^{1+\gamma} + (1 - \alpha)\theta_b \left( \frac{[\alpha\theta_a]^{\frac{1}{\gamma}}}{[(1 - \alpha)\theta_b(1 - t)]^{\frac{1}{\gamma}} + [\alpha\theta_a]^{\frac{1}{\gamma}}} \right)^{1+\gamma}.$$

*Proof.* In our collective approach to the household choice, the spouses' labor supply maximizes the family welfare. Given our assumptions of efficient allocation within the household, the division of labor supply is made in order to minimize the disutility costs of getting a given income level  $z \in \mathbb{R}_+$ . Therefore, the collective choice of labor supply is the solution to

$$\min_{z_a, z_b} \alpha \left[ \theta_a \frac{z_a^{1+\gamma}}{1 + \gamma} \right] + (1 - \alpha) \left[ \theta_b \frac{z_b^{1+\gamma}}{1 + \gamma} \right]$$

subject to

$$z_a(1 - t) + z_b = z.$$

The first-order condition for this problem is

$$\frac{\alpha \theta_a z_a^\gamma}{1 - t} = (1 - \alpha) \theta_b z_b^\gamma. \quad (\text{A.12})$$

Using  $z = z_a(1 - t) + z_b$  we have

$$z_a = z \left[ \left( \frac{\alpha}{1 - \alpha} \frac{\theta_a}{\theta_b} \frac{1}{1 - t} \right)^{\frac{1}{\gamma}} + 1 \right]^{-1} \quad \text{and} \quad (\text{A.13})$$

$$z_b = z \left[ \left( \frac{1 - \alpha}{\alpha} \frac{\theta_b}{\theta_a} (1 - t) \right)^{\frac{1}{\gamma}} + 1 \right]^{-1}. \quad (\text{A.14})$$

Substituting in the family utility function we have

$$V(x, z, \theta_a, \theta_b, \alpha, t) = \kappa(\alpha) + \ln x - \omega(\theta_a, \theta_b, \alpha, t) \frac{z^{1+\gamma}}{1 + \gamma}, \quad (\text{A.15})$$

where

$$\omega(\theta_a, \theta_b, \alpha, t) \equiv \alpha \theta_a \left[ 1 + \left( \frac{\alpha}{1 - \alpha} \frac{\theta_a}{\theta_b} \frac{1}{(1 - t)} \right)^{\frac{1}{\gamma}} \right]^{-(1+\gamma)} + (1 - \alpha) \theta_b \left[ 1 + \left( \frac{1 - \alpha}{\alpha} \frac{\theta_b}{\theta_a} (1 - t) \right)^{\frac{1}{\gamma}} \right]^{-(1+\gamma)}$$

and

$$\kappa(\alpha) = \alpha \ln \alpha + (1 - \alpha) \ln(1 - \alpha).$$

□

## B Generalizing Preferences: The Tax Perturbation Approach

In the previous sections, we were able to fully characterize the optimal tax schedules when spouses share the same iso-elastic preferences. This section uses a tax perturbation approach to provide expressions for the optimal marginal tax rate for the case in which preferences are fully general.

The first step is to compute the welfare and revenue effects of perturbing the baseline schedule in the direction of an arbitrary reform  $H : \mathbb{R}_+ \rightarrow \mathbb{R}$ , which amounts to computing its Gateaux differential. That is, consider a reform whereby the current schedule is replaced by a new one,  $\tilde{T} = T + \mu H$ , for  $\mu \in \mathbb{R}$ . When the reform  $H$  is a  $\mathcal{C}^2$  function, for small values of  $\mu$  the resulting tax schedule is a *perturbed* version of the baseline tax and lies within its neighborhood. This section restricts taxes to the class of joint taxation  $\mathcal{T}_0$ . Given a baseline tax system,  $T \in \mathcal{T}_0$ , we say that the perturbation  $H : \mathbb{R}_+ \rightarrow \mathbb{R}$  is *within the same class* if the perturbed tax schedule remains in the same class as  $T$ , that is,  $\tilde{T} = T + \mu H \in \mathcal{T}_0$  for small enough  $\mu$ . We will then say that the tax schedule is optimal within this class if there is no perturbation *within the same class*, which improves the planner's objective.<sup>30,31</sup>

We start by deriving the behavioral response for the change in a baseline tax

<sup>30</sup>There may still be  $H'$  for which  $\tilde{T}' = T + \mu H' \notin \mathcal{T}_0$ , which improves the planner's objective. In fact, in Section ?? we study reforms whose resulting perturbed classes are not within the class of joint taxation  $\mathcal{T}_0$ .

<sup>31</sup>Alves et al. (2019) show that formulae derived using variational methods may fail to characterize the optimum in models in which marginal rates of substitution are not exogenously ordered according to the unobserved heterogeneity, which typically arises in models with multidimensional heterogeneity. At issue is the implicit assumption that agents are never indifferent between two choices  $z$  and  $z'$  with  $z > z'$  and, hence, that perturbations induce no jumps. Therefore, unless we impose more structure on agents' preferences, our model is susceptible to Alves et al.'s (2019) critique.

schedule  $T : \mathbb{R}_+ \rightarrow \mathbb{R}$  (a candidate for the optimal schedule) in the direction of the reform  $H : \mathbb{R}_+ \rightarrow \mathbb{R}$ . Consider the program of a type  $\iota$  household facing a nonlinear tax schedule  $T : \mathbb{R}_+ \rightarrow \mathbb{R}$ :

$$\max_{(x,z) \in \mathbb{R}_+^2} \{V(x, z, \iota) \quad \text{s.t.} \quad x = z - T(z)\}. \quad (\text{B.1})$$

The solution to this program defines the earnings supply functional  $z_\iota : \mathcal{T} \rightarrow \mathbb{R}$ , where  $\mathcal{T}$  is the class of tax schedules being considered. We measure *behavioral responses* to the reform in the direction of  $H$  by the Gateaux derivative of the earnings supply functional, denoted by  $dz_\iota(T; H)$  (see (D.3) for the formal definition).

The overall marginal effect on welfare after a reform in the direction of  $H$  is given by

$$dW(x, z, \iota) = \partial_x W(x, z, \iota) \left[ -H(z) + \left( 1 + \frac{\partial_z W(x, z, \iota)}{\partial_x W(x, z, \iota)} \frac{1}{1 - T'(z)} \right) [1 - T'(z)] dz_\iota(T; H) \right] d\mu. \quad (\text{B.2})$$

The first term in (B.2), represents the difference between a dollar in the hands of a household and a dollar in the hands of the government. The second term captures the consequences of behavioral responses for government revenues. Typically, this second term is zero due to envelope. When there is dissonance, however, the envelope arguments normally used to capture the impact on welfare of these perturbations (e.g., [Saez \(2001\)](#)) must therefore be amended to include this term.<sup>32</sup>

Let us derive expressions for  $\partial_z W(x, z; \iota) / \partial_x W(x, z; \iota)$  and  $dz_\iota(T; h)$ . To make our point in as stark a manner as possible and to better communicate with the numerical example of Section 3, we restrict our attention to separable preferences of the form  $U(x_i, l_i) = u(x_i) - h(l_i)$  for functions  $u : \mathbb{R}_+ \rightarrow \mathbb{R}$  and  $h : \mathbb{R}_+ \rightarrow \mathbb{R}$ . We then define, with some abuse of notation and omitting parameters for shortness,  $x_a(x)$ ,  $x_b(x)$ ,  $z_a(z)$ , and  $z_b(z)$  as the solution to the family program in (2).<sup>33</sup> In this case, we can show that

$$\frac{\partial_z W(x, z, \iota)}{\partial_x W(x, z, \iota)} \frac{1}{1 - T'(z)} = - \frac{(2\alpha - 1) z'_a(z) + (1 - \alpha)}{(2\alpha - 1) x'_a(x) + (1 - \alpha)}.$$

Of course, if  $\alpha = 1/2$ , there is no dissonance and the marginal rate of substitution from the planner's perspective is equal to the marginal retention rate  $1 - T'(z)$ .

<sup>32</sup>It is easy to see the source of these differences. The first-order necessary conditions for program (B.1) is  $1 - T'(z) = -\partial_z V(x, z, \iota) / \partial_x V(x, z, \iota)$ . In a standard Mirrleesian model,  $W(x, z, \iota) = V(x, z, \iota)$ , so that  $1 - T'(z) = -\partial_z W(x, z, \iota) / \partial_x W(x, z, \iota)$ .

<sup>33</sup>Notice that separability of the utility implies separability between consumption and labor decisions. That is why we write  $x_a(\cdot)$  and  $x_b(\cdot)$  as a function only of  $x$ , and  $z_a(\cdot)$  and  $z_b(\cdot)$  as function only of  $z$ .

This is also true in the limiting case when  $x'_a(x) = z'_a(z)$ , but this result is unexpected.<sup>34</sup>

The behavioral response to a reform in the direction of  $H$ , measured by the Gateaux derivative, is given by

$$dz_{\iota}(T; H) = -\frac{\varepsilon^c(z, \iota)zH'(z) + \eta(z, \iota)H(z)}{1 - T'(z) + \varepsilon^c(z, \iota)zT''(z)}, \quad (\text{B.3})$$

where  $\varepsilon^c(z, \iota)$  is the compensated elasticity of household taxable earnings with respect to the retention rate  $1 - T'(z)$ , and  $\eta(z, \iota)$  is the income elasticity of taxable earnings of a type  $\iota$ , both of which are explicitly calculated in Appendix G.

**The relevant elasticities** For each household  $\iota$ , we follow [Jacquet et al. \(2013\)](#) and [Scheuer and Werning \(2017\)](#) in deriving the relevant elasticities of taxable income that take into account the endogeneity of marginal tax rates due to the local tax schedule curvature: uncompensated,  $\epsilon(z, \iota)$ , income,  $\eta(z, \iota)$ , and compensated,  $\varepsilon^c(z, \iota)$  (see Appendix G).

All of these elasticities are defined at the household level. However, the tax schedule should necessarily treat all families with the same total income equally. Therefore, we need to aggregate across all families. For each  $z \in \mathbb{R}_+$ , let

$$\bar{\varphi}(\iota|z) = \int_{z_{\iota}(T)=z} dF(\iota)$$

be the distribution of household types  $\iota$  choosing  $z$  at the candidate optimal tax schedule  $T$ . We can aggregate across households choosing  $z$  using  $\bar{\varphi}(\iota|z)$  and define aggregate elasticities as follows:

$$\varepsilon(z) = \int_{\iota \in \Lambda} \varepsilon(z, \iota) d\bar{\varphi}(\iota|z), \quad \eta(z) = \int_{\iota \in \Lambda} \eta(z, \iota) d\bar{\varphi}(\iota|z), \quad \text{and} \quad \varepsilon^c(z) = \int_{\iota \in \Lambda} \varepsilon^c(z, \iota) d\bar{\varphi}(\iota|z).$$

Let us denote

$$\Phi(z) = \int_{\iota \in \Lambda} d\bar{\varphi}(\iota|z)$$

as the empirical income distribution induced by the tax system, and let's assume it admits a density, which we shall denote by  $\varphi(z)$ .

As it turns out, elasticities  $\varepsilon(z)$ ,  $\eta(z)$ , and  $\varepsilon^c(z)$  are important for understanding the consequences of behavioral responses to government revenues. However, a novelty we uncover is that we need alternative elasticities to assess the welfare consequence of behavioral responses. For this, let the average welfare weight at

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<sup>34</sup>While  $x'_a(x)$  depends on the local curvature of  $u(\cdot)$  at the optimal choice  $(x_a(x), x - x_a(x))$ ,  $z'_a(z)$  depends on the local curvature of  $h(\cdot)$  at the optimal choice  $(z_a(z), z - z_a(z))$ . Hence, they will differ for almost all parameterizations.

income  $z$ ,  $g(z)$ , be defined as

$$g(z) = \int_{\boldsymbol{\iota} \in \Lambda} g(z, \boldsymbol{\iota}) d\bar{\varphi}(\boldsymbol{\iota}|z),$$

where  $g(z, \boldsymbol{\iota}) = \partial_x W(z - T(z), z, \boldsymbol{\iota})$ , which is positive given our specification of the welfare function (3). Hence, we define the welfare-weighted elasticity by

$$\bar{\varepsilon}(z) = \int_{\boldsymbol{\iota} \in \Lambda} \frac{g(z, \boldsymbol{\iota})}{g(z)} \varepsilon(z, \boldsymbol{\iota}) d\bar{\varphi}(\boldsymbol{\iota}|z),$$

with analogous definitions for  $\bar{\eta}(z)$ , the welfare-weighted income elasticity, and for  $\bar{\varepsilon}^c(z)$ , the welfare-weighted compensated elasticity of taxable earnings.<sup>35</sup>

In what follows, it is convenient to define the household  $\boldsymbol{\iota}$  *behavioral wedge*:

$$\xi(z, \boldsymbol{\iota}) \equiv \frac{\partial_z W(z - T(z), z, \boldsymbol{\iota})}{\partial_x W(z - T(z), z, \boldsymbol{\iota})} \frac{1}{1 - T'(z)}. \quad (\text{B.4})$$

Analogously to what we have done with elasticities, we define

$$\bar{\xi}(z) = \int_{\boldsymbol{\iota}} \frac{g(z, \boldsymbol{\iota})}{g(z)} \xi(z, \boldsymbol{\iota}) d\bar{\varphi}(\boldsymbol{\iota}|z),$$

the welfare-weighted average behavioral wedge.

## B.1 Optimal tax formula

We have already shown how dissonance changes the direct impact of tax reforms on welfare measured by the planner. Unlike the standard case of single households, the welfare impact of a small reform is not negligible. This happens because the social marginal value of income does not coincide with the private one. Adding the mechanical and behavioral effects on tax revenues along the lines of [Saez \(2001\)](#), we arrive in Proposition 2 at the expression for optimal marginal tax rates.

**Proposition 2.** *The first-order necessary conditions for the optimal joint tax schedule at*

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<sup>35</sup>Of course, if for all  $z$ ,  $\varepsilon(z, \boldsymbol{\iota})$  is the same for all  $\boldsymbol{\iota}$  (or if the co-variance between  $\partial W/\partial x$  and  $\varepsilon$  over the distribution of  $\boldsymbol{\iota}$  is zero), then the two concepts coincide.



the total household income level  $z \in \mathbb{R}_+$  can be written as

$$\begin{aligned} \frac{T'(z)}{1 - T'(z)} = & \frac{1}{\epsilon^c(z)} \frac{1 - \Phi(z)}{z\varphi(z)} \left\{ \int_z^{\bar{z}} \left[ 1 - g(\tilde{z}) + \frac{T'(\tilde{z})}{1 - T'(\tilde{z})} \eta(\tilde{z}) \right] \frac{\varphi(\tilde{z})}{1 - \Phi(\tilde{z})} d\tilde{z} \right. \\ & + \int_z^{\bar{z}} g(\tilde{z}) \{ [1 - \bar{\xi}(\tilde{z})] \bar{\eta}(\tilde{z}) - \zeta^{\eta, \xi}(\tilde{z}) \} \frac{\varphi(\tilde{z})}{1 - \Phi(\tilde{z})} d\tilde{z} \Big\} \\ & - \frac{g(z)}{\epsilon^c(z)} \{ [1 - \bar{\xi}(z)] \bar{\epsilon}^c(z) - \zeta^{\epsilon, \xi}(z) \}, \quad (\text{B.5}) \end{aligned}$$

where  $\Phi(z)$  is the empirical income distribution induced by the optimal tax system,  $\varphi(z)$  is its density,  $g(z)$  is the marginal social value of income in the hands of households with income  $z$ ,  $\bar{\xi}(z)$  is the behavioral wedge between the social values of income and its market price,  $\bar{\epsilon}^c(z)$  and  $\bar{\eta}^c(z)$  are welfare-adjusted average elasticities for households choosing  $z$ , and  $\zeta^{\epsilon, \xi}(z)$  and  $\zeta^{\eta, \xi}(z)$  are the welfare-adjusted covariances between these elasticities and the behavioral wedges for all households choosing  $z$ .

The first thing to note about (B.5) is that if  $\bar{\xi}(z) = 1$ , this is simply the [Piketty-Saez](#) intuitive expression for optimal taxes. The second important thing is that two types of aggregation are used to characterize optimal taxes: the regular empirical frequency-weighted elasticities ( $\epsilon(z)$ ,  $\epsilon^c(z)$ , and  $\eta(z)$ ) and the welfare-weighted elasticities ( $\bar{\epsilon}(z)$ ,  $\bar{\epsilon}^c(z)$ , and  $\bar{\eta}(z)$ ).

As pointed out by [Saez \(2001\)](#), using a perturbation approach allows us to write optimal tax formulae even when types are multidimensional, which is precisely our case here. However, with multidimensional heterogeneity, households of different types bunch at the same  $z$ , and we need to aggregate behavioral responses across these agents.<sup>36</sup> This is already in [Saez \(2001\)](#), as we can see if we assume  $\bar{\xi}(z) = 1$ . The novelty is that, whereas elasticities are only important in [Saez \(2001\)](#) because of their behavioral impact on revenues, they directly affect utility as perceived by the planner. In this case, averages must be welfare adjusted analogously to risk adjustments in asset pricing formulae.

Dissonance means that behavioral responses have first-order consequences on welfare, which vary across agents with the same earnings. The overall impact, which depends on the elasticities and the wedges, is weighted not by the relative empirical frequency of couples choosing  $z$  but by the welfare-adjusted frequencies. All elasticities pertaining to dissonance have a bar to denote this adjustment. The covariance terms that appear in the last two lines of (B.5) are also calculated under this welfare-adjusted measure.<sup>37</sup>

<sup>36</sup>According to [Saez \(2001\)](#) "The direct proof using elasticities shows that it is not necessary to introduce a uni-dimensional exogenous skill distribution to obtain formula (14) [our formula (B.5)]. Therefore, formula (14) might, in principle, be valid for any heterogeneous population as long as (...) are considered as average elasticities at income level  $z$ ."

<sup>37</sup>Covariance terms also appear in the optimal tax formulae for behavioral agents derived by

As it turns out, ridding the expression from these aggregation issues leads to a considerably more straightforward extension of optimal tax formulae. Toward this end, we assume that all elasticities are identical for agents choosing the same  $z$ . This will lead us to an ABC+D formula (Proposition 3) that is derived from (B.6) below.

**Corollary 1.** *Assume that for all  $z$ ,  $\varepsilon(z, \iota) = \varepsilon(z)$ ,  $\eta(z, \iota) = \eta(z)$ , and  $\varepsilon^c(z, \iota) = \varepsilon^c(z)$  for all  $\iota$  choosing  $z$ . Then, the first-order condition for the optimal joint tax schedule for couples at income level  $z$  in Proposition 2 can be rewritten as*

$$\begin{aligned} \frac{T'(z)}{1 - T'(z)} = \frac{1}{\varepsilon^c(z)} \frac{1 - \Phi(z)}{z\varphi(z)} \left\{ \int_z^{\bar{z}} \left[ 1 - g(\tilde{z}) + \frac{T'(\tilde{z})}{1 - T'(\tilde{z})} \eta(\tilde{z}) \right] \frac{\varphi(\tilde{z})}{1 - \Phi(\tilde{z})} d\tilde{z} \right. \\ \left. + \int_z^{\bar{z}} g(\tilde{z}) [1 - \bar{\xi}(\tilde{z})] \eta(\tilde{z}) \frac{\varphi(\tilde{z})}{1 - \Phi(\tilde{z})} d\tilde{z} \right\} - \bar{g}(z) [1 - \bar{\xi}(z)]. \quad (\text{B.6}) \end{aligned}$$

Because we have assumed that elasticities are identical for all households choosing the same  $z$ , we can drop the bar from almost all statistics and eliminate the covariance terms in (B.5). The only term for which we still make a distinction between the empirical distribution and the welfare-adjusted distribution concerns the behavioral wedge  $\bar{\xi}(z)$ .

Note how (B.6) identifies a new sufficient statistic that should be considered when evaluating optimal income taxes. It highlights that the planner may want to impose additional distortions to influence decisions taken within the household.

Equation (B.6) defines  $T'(z)/(1 - T'(z))$  implicitly. We can find an explicit solution by solving this integral equation.

**Proposition 3.** *Assume that  $\varepsilon(z, \iota) = \varepsilon(z)$ ,  $\eta(z, \iota) = \eta(z)$ , and  $\varepsilon^c(z, \iota) = \varepsilon^c(z)$  for all  $\iota$  choosing  $z$  and for all  $z$ . Then, the marginal tax rate for the optimal joint tax schedule for couples at income level  $z$  is given by*

$$\frac{T'(z)}{1 - T'(z)} = A(z)B(z)C(z) + D(z), \quad (\text{B.7})$$

where

$$\begin{aligned} A(z) = \frac{1}{\varepsilon^c(z)}, \quad B(z) = \frac{1 - \Phi(z)}{\varphi(z)z}, \quad D(z) = -g(z) [1 - \xi(z)] \\ \text{and} \quad C(z) = \int_z^{\bar{z}} \{1 - g(\tilde{z})\} \exp \left\{ \int_{\tilde{z}}^{\bar{z}} \frac{\eta(\tilde{\tilde{z}})}{\varepsilon^c(\tilde{\tilde{z}})} \frac{d\tilde{\tilde{z}}}{\tilde{\tilde{z}}} \right\} \frac{\varphi(\tilde{\tilde{z}})}{1 - \Phi(\tilde{\tilde{z}})} d\tilde{\tilde{z}}. \end{aligned}$$

Equation (B.7) has an ABC (and D) form similar to that in [Diamond \(1998\)](#), with

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[Gerritsen \(2016\)](#). [Gerritsen](#)'s representation slightly differs from ours since he does not work with a change in the measure, considering extra co-variance terms in the empirical measure.

an added  $D$  to account for a dissonance term. The term  $D(z)$  acts as a Pigouvian correction that raises marginal tax rates when  $\bar{\xi}(z) > 1$  and lowers it otherwise. Recalling that

$$\xi(z, \iota) \begin{matrix} \leq \\ \geq \end{matrix} 1 \Leftrightarrow \frac{\partial_z W(z - T(z), z, \iota)}{\partial_x W(z - T(z), z, \iota)} \begin{matrix} \leq \\ \geq \end{matrix} \frac{\partial_z V(z - T(z), z, \iota)}{\partial_x V(z - T(z), z, \iota)},$$

marginal tax rates are reduced when compared to the pure Mirrleesian optimum if, on average, family earnings  $z$  underestimate the disutility of effort as perceived by the planner and are increased otherwise.

For the case of iso-elastic preferences of Section 3,  $\xi(z, \iota)$  is independent of  $z$ .<sup>38</sup> We can therefore evaluate it separately from the rest of the program. Figure 9 shows how  $\xi$  varies with dissonance  $\alpha$  for different values of the relative productivity of spouses  $d = [w_a/w_b]^{1+\gamma}$ , where  $\gamma$  denotes the inverse of the Frisch elasticity of labor supply.

[ FIGURE 9 ABOUT HERE ]

Since the planner is utilitarian, for all values of  $d$ ,  $\xi = 1$  when  $\alpha = 1/2$ , and the Pigouvian term  $D$  vanishes. When  $\delta = 1$  (i.e., when both spouses are equally productive)  $\xi$  reaches a minimum exactly at  $\alpha = 1/2$ . This means that, for the special case in which spouses are equally productive, the Pigouvian term is always non-negative, and marginal tax rates are higher the greater the dissonance. Intuitively, the planner perceives a deviation from an equal split of effort as an inefficiency that leads to a underestimation of the welfare loss from work. It corrects it by discouraging work. For all other values of  $d$ ,  $\xi$  increases (respectively decreases) with  $\alpha$ , at  $\alpha = 1/2$ , if  $\delta < 1$  (respectively  $\delta > 1$ ). Starting from  $\alpha = 1/2$ , a slight increase in its value leads the couple to perceive a cost of working that is lower than the planner's if it is  $b$  who is productive, but leads to an overestimation if it is  $a$  who is more productive. The Pigouvian term will, in this case, be negative and the marginal tax lower than when there is no dissonance. Eventually, as dissonance becomes too large, the inefficiency perceived by the planner will dominate, and the Pigouvian term will become positive.

**Relationship between the optimal taxation formulae** Optimal tax formulae (17) could be directly derived from (B.5). First, note that under identical ln preferences, the marginal social value of resources is equal for all households earning the same  $z$ , leading us to drop all of the covariance terms and equalizing the empirical and

<sup>38</sup>Of course  $\iota$  will determine, along with the tax system, how much earnings,  $z$ , the family will generate. What is meant here is that knowledge about  $\iota$  is sufficient for determining  $\xi$ , for any tax system.

welfare-adjusted moments: the relevant expression for optimal taxes takes the explicit form (B.7). Second, separability greatly simplifies the  $C(z)$  term in (B.7) once one notes that, under separability,  $\eta(z)/\varepsilon^c(z) = d \ln u'(z - T(z))/dz$ . A somewhat tedious algebra is needed to show that the dissonance term,  $D(z)$ , becomes the term in the second line of (17). As in Saez (2001), it is comforting to check that the same formulae, albeit written in terms of primitives only, are reached using a mechanism design approach.

## C Utilitarian welfare criteria based on individuals

Assume that the planner is utilitarian across and within households. Recall that  $V_a(x, z, \theta_a, \theta_b, \alpha)$  and  $V_b(x, z, \theta_a, \theta_b, \alpha)$  are the indirect utility derived by spouses  $a$  and  $b$  in a couple  $(\theta_a, \theta_b)$  from the allocation  $(x, z)$  calculated in Lemma 4.

The planner's welfare of a given allocation  $(x, z) : [\underline{\theta}, \bar{\theta}] \times [\underline{\theta}, \bar{\theta}] \times (0, 1) \rightarrow \mathbb{R}_{++} \times Z$  is (neglecting the explicit dependence of  $(x, z)$  on types to simplify the formula):

$$\int_0^1 \int_{\underline{\theta}}^{\bar{\theta}} \int_{\underline{\theta}}^{\bar{\theta}} \left[ \frac{1}{2} V_a(x(\cdot), z(\cdot), \theta_a, \theta_b, \alpha) + \frac{1}{2} V_b(x(\cdot), z(\cdot), \theta_a, \theta_b, \alpha) \right] d\tilde{F}(\theta_a, \theta_b, \alpha).$$

where  $\tilde{F}$  denotes the distribution of the random vector  $(\theta_a, \theta_b, \alpha)$ . Given the characterization in Lemma 4 of  $V_a$  and  $V_b$ , we can re-write the welfare criteria as a function of  $(\omega, \omega_a)$ . In fact, substituting  $V_a$ ,  $V_b$  and  $\kappa(\alpha)$  we have

$$\begin{aligned} \frac{1}{2} V_a + \frac{1}{2} V_b &= \left( \frac{1}{2} - \alpha \right) \ln \frac{\alpha}{1 - \alpha} + V + \left[ \omega - \frac{\omega_a + \omega_b}{2} \right] \frac{z^{1+\gamma}}{1 + \gamma} \\ &= \left( \frac{1}{2} - \alpha \right) \ln \frac{\alpha}{1 - \alpha} + V + \left( \alpha - \frac{1}{2} \right) [\omega_a - \omega_b] \frac{z^{1+\gamma}}{1 + \gamma} \\ &= \left( \frac{1}{2} - \alpha \right) \ln \frac{\alpha}{1 - \alpha} + V + \left( \alpha - \frac{1}{2} \right) \left( \frac{\omega_a - \omega}{1 - \alpha} \right) \frac{z^{1+\gamma}}{1 + \gamma}, \end{aligned}$$

where in the second equality we substitute  $V_a$ ,  $V_b$  and  $\kappa(\alpha)$ . In the third and fifth equalities we used  $\omega = \alpha\omega_a + (1 - \alpha)\omega_b$ . Therefore, we can define the planner's social welfare function based on individuals in terms of family type and spouse  $a$  type as

$$W(x, z, \omega, \omega_a, \alpha) = \left( \frac{1}{2} - \alpha \right) \ln \frac{\alpha}{1 - \alpha} + V(x, z, \omega, \omega_a, \alpha) + \left( \alpha - \frac{1}{2} \right) \left( \frac{\omega_a - \omega}{1 - \alpha} \right) \frac{z^{1+\gamma}}{1 + \gamma}.$$

## D Proofs

*Proof of Proposition 1.* As in the proof of Lemma 5, let  $v(\omega)$  denote the utility of type  $\omega$  in the mechanism. Taking  $v(\omega)$  and  $z(\omega)$  as choice variables,  $x(\omega)$  is implicitly defined by

$$v(\omega) = \ln x(v(\omega), z(\omega), \omega) - \omega \frac{z(\omega)^{1+\gamma}}{1+\gamma}.$$

First, notice that we can rewrite the objective function as

$$\begin{aligned} \int_0^1 \int_{\underline{\omega}}^{\bar{\omega}} \int_{\underline{\omega}_a}^{\bar{\omega}_a} \left[ v(\omega) + \left( \alpha - \frac{1}{2} \right) \left( \frac{\omega_a - \omega}{1 - \alpha} \right) \frac{z(\omega)^{1+\gamma}}{1 + \gamma} \right] \hat{\psi}_c(\omega_a | \omega, \alpha) \hat{\psi}_m(\omega, \alpha) d\omega_a d\omega d\alpha = \\ \int_0^1 \int_{\underline{\omega}}^{\bar{\omega}} v(\omega) \left[ \int_{\underline{\omega}_a}^{\bar{\omega}_a} \hat{\psi}_c(\omega_a | \omega, \alpha) d\omega_a \right] \hat{\psi}_m(\omega, \alpha) d\omega d\alpha + \\ \int_0^1 \int_{\underline{\omega}}^{\bar{\omega}} \left( \alpha - \frac{1}{2} \right) \frac{z(\omega)^{1+\gamma}}{1 + \gamma} \int_{\underline{\omega}_a}^{\bar{\omega}_a} \left[ \left( \frac{\omega_a - \omega}{1 - \alpha} \right) \hat{\psi}_c(\omega_a | \omega, \alpha) d\omega_a \right] \hat{\psi}_m(\omega, \alpha) d\omega d\alpha, \end{aligned}$$

using

$$\int_{\underline{\omega}_a}^{\bar{\omega}_a} \hat{\psi}_c(\omega_a | \omega, \alpha) d\omega_a = 1$$

and

$$\int_{\underline{\omega}_a}^{\bar{\omega}_a} \omega_a \hat{\psi}_c(\omega_a | \omega, \alpha) d\omega_a \equiv \mathbb{E}[\omega_a | \omega, \alpha].$$

Define the marginal density

$$\psi(\omega) \equiv \int_0^1 \hat{\psi}_m(\omega, \alpha) d\alpha.$$

We can rewrite the objective function as

$$\int_{\underline{\omega}}^{\bar{\omega}} v(\omega) \psi(\omega) d\omega + \int_{\underline{\omega}}^{\bar{\omega}} \int_0^1 \left[ \left( \alpha - \frac{1}{2} \right) \left( \frac{\mathbb{E}[\omega_a | \omega, \alpha] - \omega}{1 - \alpha} \right) \right] \frac{z(\omega)^{1+\gamma}}{1 + \gamma} \hat{\psi}_m(\omega, \alpha) d\alpha d\omega.$$

Thus, the planner chooses  $v, z : \Omega \rightarrow \mathbb{R}$  to maximize this objective function subject to the local first-order local incentive constraint:

$$\dot{v}(\omega) = \frac{z(\omega)^{1+\gamma}}{1 + \gamma}; \tag{D.1}$$

and monotonicity constraints:  $z(\omega)$  is increasing; and budget constraint:

$$\int_{\underline{\omega}}^{\bar{\omega}} [z(\omega) - x(v(\omega), z(\omega), \omega)] \psi(\omega) d\omega \geq 0.$$

By the implicit function theorem,

$$x_v = \frac{1}{u'(x(v(\omega), z(\omega), \omega))} = x(v(\omega), z(\omega), \omega),$$

$$x_z = \frac{h'(z(\omega))}{u'(x(v(\omega), z(\omega), \omega))} = \omega z(\omega)^\gamma x(v(\omega), z(\omega), \omega).$$

In this approach the income  $z(\omega)$  is the control variable and utility of the family  $v(\omega)$  is the state variable.

The Hamiltonian associated to this problem is

$$\mathcal{H}(v(\omega), z(\omega), \mu(\omega), \lambda, \omega) = \frac{z(\omega)^{1+\gamma}}{1+\gamma} \left[ \int_0^1 \left( \alpha - \frac{1}{2} \right) \left( \frac{\mathbb{E}[\omega_a | \omega, \alpha] - \omega}{1-\alpha} \right) \hat{\psi}_m(\omega, \alpha) d\alpha \right]$$

$$+ v(\omega) \psi(\omega) + \lambda [z(\omega) - x(v(\omega), z(\omega), \omega)] \psi(\omega) - \mu(\omega) \frac{z(\omega)^{1+\gamma}}{1+\gamma},$$

where  $\lambda \in \mathbb{R}_+$  is the Lagrange multiplier associated to the government budget constraint, and  $\mu(\omega)$  is the co-state variable. At the optimum, the control  $z(\omega)$  maximizes the Hamiltonian function giving the following first-order condition:

$$z(\omega)^\gamma \left[ \int_0^1 \left( \alpha - \frac{1}{2} \right) \left( \frac{\mathbb{E}[\omega_a | \omega, \alpha] - \omega}{1-\alpha} \right) \hat{\psi}_m(\omega, \alpha) d\alpha \right] + \lambda(1 - x_z) \psi(\omega) = \mu(\omega) z(\omega)^\gamma$$

and the other first-order condition of the Hamiltonian is

$$\dot{\mu}(\omega) = -\frac{\partial}{\partial v} \mathcal{H}(\cdot) = -[1 - \lambda x_v] \psi(\omega)$$

or

$$\dot{\mu}(\omega) = \psi(\omega) [\lambda x_v - 1]$$

and the transversality conditions  $\mu(\underline{\omega}) = \mu(\bar{\omega}) = 0$ . Integrating on both sides from  $\omega$  to  $\bar{\omega}$  and applying the fundamental theorem of calculus we have

$$\mu(\bar{\omega}) - \mu(\omega) = \int_\omega^{\bar{\omega}} \dot{\mu}(s) ds = \int_\omega^{\bar{\omega}} \psi(s) [\lambda x_v - 1] ds.$$

Using the transversality condition we have

$$\mu(\omega) = \lambda \int_\omega^{\bar{\omega}} \psi(s) [\lambda^{-1} - x_v(v(s), z(s), s)] ds. \quad (\text{D.2})$$

Note that from the transversality condition  $\mu(\underline{\omega}) = 0$  we can pin down the social

value of an extra unit of income:

$$0 = \mu(\underline{\omega}) = \lambda \int_{\underline{\omega}}^{\bar{\omega}} \psi(s) \left[ \frac{1}{\lambda} - x_v(v(s), z(s), s) \right] ds.$$

Therefore,

$$\lambda = \frac{1}{\int_{\underline{\omega}}^{\bar{\omega}} x_v(v(s), z(s), s) \psi(s) ds} = \frac{1}{\int_{\underline{\omega}}^{\bar{\omega}} x(v(s), z(s), s) \psi(s) ds}$$

or

$$\lambda = \mathbb{E}^\psi[x]^{-1}.$$

Substituting  $\mu(\omega)$  in (D.2) into equation (D.2) we have

$$\begin{aligned} z(\omega)^\gamma \left[ \int_0^1 \left( \alpha - \frac{1}{2} \right) \left( \frac{\mathbb{E}[\omega_a|\omega, \alpha] - \omega}{1 - \alpha} \right) \hat{\psi}_m(\omega, \alpha) d\alpha \right] + \lambda(1 - x_z)\psi(\omega) = \\ z(\omega)^\gamma \lambda \int_{\omega}^{\bar{\omega}} \psi(s) [\lambda^{-1} - x_v(v(s), z(s), s)] ds. \end{aligned}$$

This gives us the following expression for the marginal tax rate

$$\begin{aligned} T'(z(\omega)) = \frac{z(\omega)^\gamma}{\psi(\omega)} \int_{\omega}^{\bar{\omega}} \left[ x_v(v(s), z(s), s) - \frac{1}{\lambda} \right] \psi(s) ds \\ - \frac{z(\omega)^\gamma}{\lambda \psi(\omega)} \left[ \int_0^1 \left( \alpha - \frac{1}{2} \right) \left( \frac{\mathbb{E}[\omega_a|\omega, \alpha] - \omega}{1 - \alpha} \right) \hat{\psi}_m(\omega, \alpha) d\alpha \right]. \end{aligned}$$

Using  $\lambda = \mathbb{E}^\psi[x]^{-1}$  and  $x_v = x(v(\omega), z(\omega), \omega)$ , we can rewrite with some abuse of notation ( $x_v(v(s), z(s), s) = x(s)$ ) the formula for the optimal marginal taxation as

$$\begin{aligned} T'(z(\omega)) = \frac{z(\omega)^\gamma}{\psi(\omega)} \int_{\omega}^{\bar{\omega}} [x(s) - \mathbb{E}^\psi[x]] \psi(s) ds \\ - \frac{z(\omega)^\gamma}{\psi(\omega)} \mathbb{E}^\psi[x] \left[ \int_0^1 \left( \alpha - \frac{1}{2} \right) \left( \frac{\mathbb{E}[\omega_a|\omega, \alpha] - \omega}{1 - \alpha} \right) \hat{\psi}_m(\omega, \alpha) d\alpha \right]. \end{aligned}$$

For future reference if we assume  $\alpha$  to be constant across all households, this last formula simplifies to

$$T'(z(\omega)) = \frac{z(\omega)^\gamma}{\psi(\omega)} \int_{\omega}^{\bar{\omega}} [x(s) - \mathbb{E}^\psi[x]] \psi(s) ds - z(\omega)^\gamma \mathbb{E}^\psi[x] \left( \alpha - \frac{1}{2} \right) \left( \frac{\mathbb{E}[\omega_a|\omega] - \omega}{1 - \alpha} \right).$$

□

*Proof of Proposition 2.* The Gateaux derivative of the earning supply functional, is

defined by

$$dz_{\iota}(T; H) \equiv \lim_{\mu \rightarrow 0} \frac{z_{\iota}(T + \mu H) - z_{\iota}(T)}{\mu} = \left. \frac{\partial z_{\iota}}{\partial \mu} \right|_{\mu=0}. \quad (\text{D.3})$$

Increase the marginal tax rate by  $d\tau$  in a small interval  $(z', z' + dz')$ . Let  $\Phi(z)$  denote the distribution of income induced by the candidate optimal tax schedule,  $T(\cdot)$  (and the distribution  $F(\cdot)$  of types  $\iota$ ). Let us then consider the overall impact of such a reform.

**Impact on government revenues** For agents with taxable income in the interval  $(z', z' + dz')$ , there is a mechanical effect  $M_1 = d\tau dz'$ . But there is also a behavioral effect, which is given by

$$B_1(z', \iota) = T'(z') \left. \frac{dz}{d\tau} \right|_{\iota} d\tau = T'(z') \epsilon^c(z', \iota) \frac{z'}{1 - T'(z')} d\tau.$$

Let  $\bar{\varphi}(\iota|z')$  be the conditional distribution on  $z'$  of  $\iota$ . Then, the average response at income  $z'$  is

$$B_1(z') = \frac{z' T'(z')}{1 - T'(z')} \int \epsilon^c(z', \iota) d\bar{\varphi}(\iota|z') d\tau = \frac{z' T'(z')}{1 - T'(z')} \epsilon^c(z') d\tau,$$

for  $\epsilon^c(z')$  as defined in Section B.

This affects agents in the interval  $(z', z' + dz')$ , which results in an overall effect

$$\epsilon^c(z') \frac{T'(z')}{1 - T'(z')} z' \varphi(z') dz' d\tau,$$

where  $\varphi$  is the density of  $z$  induced by the tax system.

For agents with taxable income in  $[z', \bar{z}]$ , we have again a mechanical effect  $d\tau dz'$  which affects all agents for an overall increase in revenues given by

$$M_2 = [1 - \Phi(z')] d\tau dz'.$$

For all agents in this interval there is also a behavioral response due to income effect

$$B_2(z, \iota) = T'(z) \frac{dz}{dI} d\tau dz' = \frac{T'(z)}{1 - T'(z)} \eta(z, \iota) d\tau dz',$$

which can be aggregated across all households earning  $z$  through

$$B_2(z) = \frac{T'(z)}{1 - T'(z)} \int \eta(z, \iota) d\bar{\varphi}(\iota|z) d\tau dz' = \frac{T'(z)}{1 - T'(z)} \eta(z) d\tau dz'.$$

Finally, we can use the expression above to find the overall behavior effect on tax



revenues for agents in this range

$$B_2(z, \boldsymbol{\iota}) = d\tau dz' \int_{z'}^{\bar{z}} \frac{T'(z)}{1 - T'(z)} \eta(z) \varphi(z) dz.$$

This is exactly as in [Saez \(2001\)](#) with the proviso that we are using a different definition for the elasticities we are aggregating.

**Welfare impacts** As for the welfare effects, again, let us start with households with taxable income in the interval  $(z', z' + dz')$ . First note that for this group, the tax reform has no first-order mechanical effect on utility,  $V$ . The relevant ‘decision’ elasticities are, therefore, Hicksian.

The impact on welfare is however measured according to the planner’s valuation, i.e., the ‘experience’ utility  $W$ . Hence,

$$dW_1(z', \boldsymbol{\iota}) = \frac{\partial}{\partial x} W(z' - T(z'), z', \boldsymbol{\iota}) [1 - T'(z')] \left. \frac{dz}{d\tau} \right|_u d\tau + \frac{\partial}{\partial z} W(z' - T(z'), z', \boldsymbol{\iota}) \left. \frac{dz}{d\tau} \right|_u d\tau,$$

which simplifies to

$$dW_1(z', \boldsymbol{\iota}) = \frac{\partial}{\partial x} W(z' - T(z'), z', \boldsymbol{\iota}) \left[ 1 - T'(z') + \frac{\frac{\partial}{\partial z} W(z' - T(z'), z', \boldsymbol{\iota})}{\frac{\partial}{\partial x} W(z' - T(z'), z', \boldsymbol{\iota})} \right] \left. \frac{dz}{d\tau} \right|_u d\tau,$$

or

$$dW_1(z', \boldsymbol{\iota}) = \frac{\partial}{\partial x} W(z' - T(z'), z', \boldsymbol{\iota}) [1 - \xi(z', \boldsymbol{\iota})] \epsilon^c(z', \boldsymbol{\iota}) z' d\tau,$$

where, for all  $z$ ,

$$\xi(z, \boldsymbol{\iota}) \equiv \frac{1}{1 - T'(z)} \frac{\frac{\partial}{\partial z} W(z - T(z), z, \boldsymbol{\iota})}{\frac{\partial}{\partial x} W(z - T(z), z, \boldsymbol{\iota})}.$$

Let  $\bar{\varphi}(\boldsymbol{\iota}|z')$  the conditional on  $z'$  distribution of  $\boldsymbol{\iota}$ . The overall welfare effect at  $z'$  is

$$W_1 = dz' d\tau z' \int [g(z', \boldsymbol{\iota}) [1 - \xi(z', \boldsymbol{\iota})] \epsilon^c(z', \boldsymbol{\iota})] d\bar{\varphi}(\boldsymbol{\iota}|z'),$$

for  $g(z', \boldsymbol{\iota})$  the marginal social value of income in the hands of a  $\boldsymbol{\iota}$  household earning  $z'$ .

To simplify this expression, let  $\tilde{\varphi}(\cdot|z)$  be a measure equivalent to  $\bar{\varphi}(\cdot|z)$ , ( $\tilde{\varphi}(\cdot|z) \equiv \bar{\varphi}(\cdot|z)$ ) whose Radon-Nikodym derivative with respect to  $\bar{\varphi}(\cdot|z')$  is the function  $g(z, \cdot)/\bar{g}(z)$ , where  $\bar{g}(z) = \int g(z, \boldsymbol{\iota}) d\bar{\varphi}(\boldsymbol{\iota}|z)$ . This measure allows us to re-express the definitions for  $\bar{\xi}(z)$  and  $\bar{\epsilon}^c(z)$  from Section B, as  $\bar{\xi}(z) = \int \xi(z, \boldsymbol{\iota}) d\tilde{\varphi}(\boldsymbol{\iota}|z) = \mathbb{E}[\xi(z, \boldsymbol{\iota})]$ ,

and  $\bar{\varepsilon}^c(z) = \tilde{\mathbb{E}}[\varepsilon^c(z, \boldsymbol{\iota})]$ . It also allows us to define a covariance term

$$\zeta^{\varepsilon, \xi}(z) = c\tilde{o}v\left(\varepsilon^c(z, \boldsymbol{\iota}), \xi(z, \boldsymbol{\iota})\right) = \tilde{\mathbb{E}}[\varepsilon^c(z, \boldsymbol{\iota})\xi(z, \boldsymbol{\iota})] - \bar{\xi}(z)\bar{\varepsilon}^c(z).$$

Using these definitions we write

$$W_1 = dz'd\tau z' \bar{g}(z') \left[ [1 - \bar{\xi}(z')] \bar{\varepsilon}^c(z') - \zeta^{\varepsilon, \xi}(z') \right].$$

For households with taxable income  $z \geq z'$ ,

$$\begin{aligned} dW_2(z, \boldsymbol{\iota}) = & -\frac{\partial}{\partial x} W(z - T(z), z, \boldsymbol{\iota}) dz' d\tau + \\ & \frac{\partial}{\partial x} W(z - T(z), z, \boldsymbol{\iota}) [1 - T'(z)] \frac{dz}{dI} dz' d\tau + \frac{\partial}{\partial z} W(z - T(z), z, \boldsymbol{\iota}) \frac{dz}{dI} dz' d\tau \end{aligned}$$

leading to

$$dW_2(z, \boldsymbol{\iota}) = -\frac{\partial}{\partial x} W(z - T(z), z, \boldsymbol{\iota}) \left\{ 1 + \left[ 1 - T'(z) + \frac{\frac{\partial}{\partial z} W(z - T(z), z, \boldsymbol{\iota})}{\frac{\partial}{\partial x} W(z - T(z), z, \boldsymbol{\iota})} \right] \frac{dz}{dI} \right\} dz' d\tau,$$

and finally

$$dW_2(z, \boldsymbol{\iota}) = -g(z, \boldsymbol{\iota}) \{1 + [1 - \xi(z, \boldsymbol{\iota})] \eta(z, \boldsymbol{\iota})\} dz' d\tau.$$

To aggregate across all household earnings at the same  $z$ , we integrate over  $\boldsymbol{\iota}$

$$d\bar{W}_2(z) = dz' d\tau \int \left[ -g(z, \boldsymbol{\iota}) \{1 + [1 - \xi(z, \boldsymbol{\iota})] \eta(z, \boldsymbol{\iota})\} \right] d\bar{\varphi}(\boldsymbol{\iota}|z').$$

Analogously, to what we have done for  $dW_1$  we can further simplify this expression to

$$d\bar{W}_2(z) = d\tau dz' \bar{g}(z) \left[ [1 - \bar{\xi}(z)] \bar{\eta}^c(z) - \zeta^{\eta, \xi}(z) \right],$$

where the covariance term  $\zeta^{\eta, \xi}(z)$  is also under the equivalent measure,  $\tilde{\varphi}(\cdot|z)$ .

The overall effect on this household welfare is

$$W_2 = d\tau dz' \int_{z'}^{\bar{z}} \bar{g}(z) \{1 + [1 - \bar{\xi}(z)] \bar{\eta}(z) - \zeta^{\eta, \xi}(z)\} \varphi(z) dz.$$

At the optimum all these effects must cancel out and we get

$$M_1 + M_2 + B_1 + B_2 + W_1 + W_2 = 0.$$

Substituting the expressions for each on of these terms we get

$$\begin{aligned} & \frac{T'(z')}{1 - T'(z')} \epsilon^c(z') z' \varphi(z') - g(z') \left[ [1 - \xi(z')] \bar{\epsilon}^c(z') - \zeta^{\epsilon, \xi}(z') \right] z' \varphi(z') + [1 - \Phi(z')] \\ & + \int_{z'}^{\bar{z}} \frac{T'(z)}{1 - T'(z)} \eta(z) \varphi(z) dz + \int_{z'}^{\bar{z}} g(z) \{ 1 + [1 - \xi(z)] \bar{\eta}(z) - \zeta^{\eta, \xi}(z) \} \varphi(z) dz = 0 \end{aligned}$$

Reorganizing the terms,

$$\begin{aligned} & \frac{T'(z')}{1 - T'(z')} \epsilon^c(z') z' \varphi(z') = \int_{z'}^{\bar{z}} \left[ 1 - g(z) + \frac{T'(z)}{1 - T'(z)} \eta(z) \right] \varphi(z) dz \\ & + \int_{z'}^{\bar{z}} g(z) \{ [1 - \xi(z)] \bar{\eta}(z) - \zeta^{\eta, \xi}(z) \} \varphi(z) dz \\ & + g(z') \left[ [1 - \xi(z')] \bar{\epsilon}^c(z') - \zeta^{\epsilon, \xi}(z') \right] z' \varphi(z') \end{aligned}$$

or

$$\begin{aligned} \frac{T'(z)}{1 - T'(z)} &= \frac{1}{\epsilon^c(z)} \frac{1 - \Phi(z)}{z \varphi(z)} \left\{ \int_z^{\bar{z}} \left[ 1 - g(\tilde{z}) + \frac{T'(\tilde{z})}{1 - T'(\tilde{z})} \eta(\tilde{z}) \right] \frac{\varphi(\tilde{z})}{1 - \Phi(\tilde{z})} dz \right. \\ & + \int_{z'}^{\bar{z}} g(\tilde{z}) \{ [1 - \xi(\tilde{z})] \bar{\eta}(\tilde{z}) - \zeta^{\eta, \xi}(\tilde{z}) \} \frac{\varphi(\tilde{z})}{1 - \Phi(\tilde{z})} dz \Big\} \\ & - \frac{g(z)}{\epsilon^c(z)} \{ [1 - \xi(z)] \bar{\epsilon}^c(z) - \zeta^{\epsilon, \xi}(z) \} \quad (\text{D.4}) \end{aligned}$$

□

*Proof of Proposition 1.* We prove Proposition 1 as a particular case of Proposition 2. Under the assumptions of Proposition 1,  $\xi(z) = \bar{\xi}(z)$ ,  $\epsilon^c(z) = \bar{\epsilon}^c(z)$  and  $\eta(z) = \bar{\eta}(z)$ , for all  $z$ . Moreover,  $\zeta^{\epsilon, \xi}(z) = 0$  and  $\zeta^{\eta, \xi}(\tilde{z}) = 0$ , for all  $z$ . As a consequence, expression (B.5) collapses to (B.6).

If we define

$$\kappa(z) := \frac{T'(z)}{1 - T'(z)},$$

$$A(z)^{-1} := \frac{1}{\epsilon^c(z)} \frac{1 - \Phi(z)}{z \varphi(z)},$$

$$B(z) := [1 - g(z) + g(z)[1 - \xi(z)]\eta(z)] \frac{\varphi(\tilde{z})}{1 - \Phi(\tilde{z})},$$

$$C(z) := \eta(z) \frac{\varphi(\tilde{z})}{1 - \Phi(\tilde{z})}$$

and

$$D(z) = g(z)[1 - \xi(z)]A(z),$$

then (D.4) can be written as

$$A(z)\kappa(z) = \int_z^{\bar{z}} [B(\tilde{z}) + C(\tilde{z})\kappa(\tilde{z})] d\tilde{z} - D(z)$$

and solved for

$$\kappa(z) = -\frac{1}{A(z)} \int_z^{\bar{z}} \left[ B(\tilde{z}) + \frac{D(\tilde{z})C(\tilde{z})}{A(\tilde{z})} - D(\tilde{z}) \right] \exp \left\{ \int_z^{\tilde{z}} \frac{C(\tilde{\tilde{z}})}{A(\tilde{\tilde{z}})} d\tilde{\tilde{z}} \right\} d\tilde{z} + \frac{D(z)}{A(z)}$$

or

$$\begin{aligned} \frac{T'(z)}{1 - T'(z)} = \frac{1}{\varepsilon^c(z)} \frac{1 - \Phi(z)}{\varphi(z)z} \int_z^{\bar{z}} \{g(\tilde{z}) - 1\} \exp \left\{ \int_z^{\tilde{z}} \frac{\eta(\tilde{\tilde{z}})}{\varepsilon^c(\tilde{\tilde{z}})} \frac{d\tilde{\tilde{z}}}{\tilde{\tilde{z}}} \right\} \frac{\varphi(\tilde{z})}{1 - \Phi(z)} d\tilde{z} \\ - g(z) \{1 - \xi(z)\}. \end{aligned} \quad (\text{D.5})$$

□

## E Implementation Algorithm for the Numeric Simulations

In this section we compute numerically the optimal tax schedules derived in Proposition 1 and Corollary ??, using real data from the 2017 March CPS to have a clear idea of the shape of the optimal taxation in both cases. We use the sample of couples without dependents, using wages as a proxy for types.

Note that the marginal tax rate in Proposition 1 depends on the distribution of  $\omega$ ,  $\psi(\omega)$ , and on the distribution of  $\omega_a$ , conditional on  $\omega$ ,  $\phi(\omega_a | \omega)$ . However, we can calculate the implied distribution of  $\omega$ ,  $\psi(\omega)$ , from the knowledge of the wage distribution. It is important to notice that we only need to estimate the conditional expectation  $\mathbb{E}[\omega_a | \omega]$  and we do that non-parametrically using a simple kernel regression. This simplifies our analysis by avoiding the calculation of the conditional distribution of  $\omega$ ,  $\phi(\omega_a | \omega)$  without relying on any additional restrictive assumptions.

After calculating these objects, we simulate the marginal tax rate by adapting the algorithm proposed by [Mankiw et al. \(2009\)](#), based on a discrete grid of the empirical distributions. The algorithm goes as in the following steps:

*Step 1:* Given  $\psi(\omega)$ , discretize the probability mass function  $\pi(\omega)$  in the following way: divide the interval considered  $[\underline{\omega}, \bar{\omega}]$  in  $N$  bins of equal bandwidth  $\Delta$ , creating a grid where  $\{\omega_i\}_{i=1}^N$  with  $\underline{\omega} \leq \omega_1 \leq \dots \leq \omega_n \leq \dots \leq \omega_N \leq \bar{\omega}$  are the

midpoints in the grid. The probability mass function at each  $\omega$  is given by

$$\pi(\omega) = \Psi(\omega + \Delta/2) - \Psi(\omega - \Delta/2),$$

where  $\Psi(\cdot)$  is the cumulative distribution function of  $\omega$ .

*Step 2:* For each  $\{\omega_i\}_{i=1}^N$  estimate the conditional expectation using the Nadaraya and Watson kernel regression as

$$\hat{\mathbb{E}}[\omega_a|\omega_i] = \frac{1/(Nh) \sum_{j=1}^N K(\frac{\omega_j - \omega_i}{h}) \omega_{aj}}{1/(Nh) \sum_{j=1}^N K(\frac{\omega_j - \omega_i}{h})},$$

where  $K(\cdot)$  is the Gaussian density and  $h$  is the bandwidth.

Now we start the loop for the calculation of the optimal marginal tax rate  $T_k(\omega_i)$ .

*Step 3:* Start with a simple guess for the marginal tax rate. For instance, a flat tax schedule  $T_0(\omega_i) = .35$  for all  $\omega_i$  in the grid and a lump-sum transfer  $t_0 = .0001$ .

*Step 4:* Since  $T_{k-1}(\cdot)$  at each iteration is only calculated at points of the grid, extrapolate  $T_{k-1}(\cdot)$  to be defined as well at points outside the grid in the following way:

$$T'_{k-1}(\omega) = \begin{cases} T'_{k-1}(\omega_1), & \text{for } \underline{\omega} \leq \omega \leq \omega_1; \\ T'_{k-1}(\omega_2), & \text{for } \omega_1 \leq \omega \leq \omega_2; \\ \dots & \\ T'_{k-1}(\omega_n), & \text{for } \omega_{n-1} \leq \omega \leq \omega_n; \\ \dots & \\ T'_{k-1}(\omega_N), & \text{for } \omega_{N-1} \leq \omega \leq \bar{\omega}; \end{cases}$$

generating the tax schedule

$$T_{k-1}(\omega_i) = \begin{cases} T'_{k-1}(\omega_1)z(\omega_i) - t_{k-1}, & \text{for } \underline{\omega} \leq \omega_i \leq \omega_1; \\ T'_{k-1}(\omega_1)z(\omega_1) + T'_{k-1}(\omega_2)[z(\omega_i) - z(\omega_1)] - t_{k-1}, & \text{for } \omega_1 \leq \omega_i \leq \omega_2; \\ \dots & \\ T'_{k-1}(\omega_1)z(\omega_1) + \sum_{j=2}^{n-1} T'_{k-1}(\omega_j)[z(\omega_j) - z(\omega_{j-1})] + \\ T'_{k-1}(\omega_n)[z(\omega_i) - z(\omega_{n-1})] - t_{k-1}, & \text{for } \omega_{n-1} \leq \omega_i \leq \omega_n; \end{cases}$$

*Step 5:* Given  $T_{k-1}(\cdot)$ , calculate the optimal labor supply at each  $\omega_i$  by solving

$$\max_z \ln(z - T_{k-1}(z)) - \omega_i \frac{z^{1+\gamma}}{1+\gamma}.$$

The necessary first-order condition for the type  $\omega_i$  problem is

$$1 - T'_{k-1}(z) = \omega_i z^\gamma (z - T_{k-1}(z)) \quad (\text{E.1})$$

and the optimal  $z^*(\omega_i)$  is implicitly defined in this first-order condition.

*Step 6:* Given the optimal choice of  $z(\omega_i)$ , we can define the consumption as

$$x(\omega_i) = z^*(\omega_i) - T_{k-1}(z^*(\omega_i))$$

and use all the information to update the marginal tax rate in each point of the grid using Corollary ?? and the first-order condition of the planner's problem as follows

$$T'_k(\omega_i) = \frac{z^*(\omega_i)^\gamma}{\psi(\omega_i)} \int_{\omega_i}^{\bar{\omega}} [x(s) - \mathbb{E}^\psi[x]] \psi(s) ds - z^*(\omega_i)^\gamma \mathbb{E}^\psi[x] \left( \alpha - \frac{1}{2} \right) \left( \frac{\mathbb{E}[\omega_a|\omega_i] - \omega_i}{1 - \alpha} \right). \quad (\text{E.2})$$

Since the family utility satisfies the single-crossing property w.r.t.  $\omega_i$ , this first-order condition and the monotonicity of  $z^*(\omega_i)$  are sufficient for optimality.

*Step 7:* The formula in *Step 6* is unfeasible to estimate given our approximation of the wage distribution in the grid. Therefore, we must use a discrete approximation. One possibility is as follows:

$$\begin{aligned} T'_k(\omega_i) &\approx \frac{1}{x(\omega_i)\omega_i} \frac{1}{\pi(\omega_i)/\Delta} \left[ \sum_{j=i+1}^N [x(\omega_j) - \hat{\mathbb{E}}[x]] \pi(\omega_j) \right] \\ &\quad - \frac{1}{x(\omega_i)\omega_i} \hat{\mathbb{E}}[x] \left( \alpha - \frac{1}{2} \right) \left( \frac{\hat{\mathbb{E}}[\omega_a|\omega_i] - \omega_i}{1 - \alpha} \right) \\ &= \frac{1}{x(\omega_i)\omega_i} \frac{1}{\pi(\omega_i)/\Delta} \left[ \sum_{j=i+1}^N x(\omega_j) \pi(\omega_j) - \hat{\mathbb{E}}[x] (1 - \Pi(\omega_i)) \right] \\ &\quad - \frac{1}{x(\omega_i)\omega_i} \hat{\mathbb{E}}[x] \left( \alpha - \frac{1}{2} \right) \left( \frac{\hat{\mathbb{E}}[\omega_a|\omega_i] - \omega_i}{1 - \alpha} \right), \quad (\text{E.3}) \end{aligned}$$

where  $\hat{\mathbb{E}}[\omega_a|\omega_i]$  is calculated at *Step 2* and

$$\hat{\mathbb{E}}[x] = \sum_{j=1}^N x(\omega_j) \pi(\omega_j).$$

*Step 8:* Update  $t$  to ensure the budget constraint in the following way:

$$t_k = \sum_{j=1}^N T_k(\omega_j) \pi(\omega_j).$$

*Step 9:* Return to *Step 4* until

$$\max_i \{|T'_k(\omega_i) - T'_{k-1}(\omega_i)|\} < 10^{-3}.$$

*Step 10:* After getting a fixed-point of this operator, the only thing to be checked is the monotonicity to guarantee that the found tax schedule is implementable.

Note that this approximation is increasingly precise as  $\Delta \rightarrow 0$  and  $\omega_N \rightarrow +\infty$ . From this numerical exercise, we can make some counter-factuals. For example, evaluating how the marginal tax rates vary as the spouses' bargaining power changes within the household.

## F The role of $\alpha$

To assess the sensitivity of the optimal policy to  $\alpha$ , the parameter measuring power, we recalculate the optimal tax schedule under three alternative assumptions regarding  $\alpha$ . For  $\alpha = 0.3$ , we make power even more biased towards the primary earner. For  $\alpha = 0.7$ , we ask how policy changes when the least productive agent has on average more power. Finally, we use  $\alpha = 0.5$  to show how our model reproduces the classic findings in optimal taxation when there is no misalignment between the planner and the households.

The  $\alpha = 0.3$  scenario corresponds to the situation in which the female spouse has a low weight on the family utility function and gets a small share of family resources for her private consumption. The  $\alpha = 0.7$  is the opposite scenario in which the male spouse has a low weight and a small share of family resources. In the  $\alpha = 0.5$  scenario, resources are equally distributed between the spouses, and this is the implicit assumption made whenever a household-oriented approach is taken. Under this assumption, the marginal tax rate in Corollary ?? depends on the distributions of the family unitary type  $\omega$ , its distribution  $\psi(\omega)$ , as well as the distribution of the modified type  $\omega_a$  conditional on  $\omega$ ,  $\phi(\omega_a | \omega)$ . We calculate these distributions from the knowledge of the empirical wage distribution.

Figure 10 displays the relationship of female adjusted types  $\omega_a$  and the family unitary type  $\omega$ . It displays a scatter plot of the families in our sample along with the estimated conditional expectation  $\mathbb{E}[\omega_a | \omega]$  (dashed blue line) implemented non-parameterically using kernel regression.

[ FIGURE 10 ABOUT HERE ]

In Figure 10, we can see how the adjusted types change with  $\alpha$ . The intuition is that types are adjusted to compensate for the bargaining power within the family. For instance, adjusted types work as if we boost types for spouses with a smaller Pareto weight in the reduced-form utility.

Comparing the three panels in Figure 10, we see that the impact of collective decision making can be empirically relevant. As  $\alpha$  increases,  $\mathbb{E}[\omega_a|\omega]$  decreases. An interpretation is that the greater the woman's welfare weight for a given family type, the more her welfare relies on it and the less it relies on her original productivity: a greater fraction of household earnings is assigned to the husband. Hence, we expect that decreasing the woman's adjusted type will be needed.

[ FIGURE 11 ABOUT HERE ]

[ FIGURE 12 ABOUT HERE ]

Figure 12 highlights the the Pigouvian correction introduced by dissonance. This term can be negative, and its absolute value can be bigger than the standard term. When this happens, it leads to a marginal subsidy. This marginal subsidy is a new feature of optimal marginal tax rates. Typically, the modified term has a U-shaped format, starting high for low values of the family unitary type  $\omega$  and steadily decreasing up to a point where it becomes negative.

## G Elasticities

We first recall that in [Saez's](#) original paper, elasticities were defined for linearized budget sets. Here, we follow, instead, [Jacquet et al. \(2013\)](#) and [Scheuer and Werning \(2017\)](#) in using the elasticities defined under nonlinear budget sets. That is, we define the elasticity of taxable income with respect to the retention rate  $1 - T'(z)$ ,  $\epsilon(z, \mathbf{t})$ , as

$$\epsilon(z, \mathbf{t}) = \Delta^{-1} \left[ \partial_{xz} V(x, z, \mathbf{t}) z - \partial_{xx} V(x, z, \mathbf{t}) \frac{\partial_z V(x, z, \mathbf{t})}{\partial_x V(x, z, \mathbf{t})} z - \partial_x V(x, z, \mathbf{t}) \right] \frac{1 - T'(z)}{z},$$

where

$$\Delta = \left[ \partial_{xx} V(x, z, \mathbf{t}) \left( \frac{\partial_z V(x, z, \mathbf{t})}{\partial_x V(x, z, \mathbf{t})} \right)^2 - 2 \partial_{xz} V(x, z, \mathbf{t}) \frac{\partial_z V(x, z, \mathbf{t})}{\partial_x V(x, z, \mathbf{t})} + \partial_{zz} V(x, z, \mathbf{t}) - \partial_x V(x, z, \mathbf{t}) T''(z) \right].$$



The last term in  $\Delta$ ,  $\partial_x V(x, z)T''(z)$ , captures the curvature in the budget set. Saez (2001), in contrast, defines the elasticity using  $\tilde{\Delta} = \Delta + \partial_x V(x, z)T''(z)$ . The advantage of using  $\tilde{\Delta}$  is its familiarity, whereas the advantage of using  $\Delta$  is the simplification of optimal tax formulae.

We can also define the income elasticity of taxable earnings

$$\eta(z, \boldsymbol{\iota}) = \Delta^{-1} \left[ \partial_{xx} V(x, z, \boldsymbol{\iota}) \frac{-\partial_z V(x, z, \boldsymbol{\iota})}{\partial_x V(x, z, \boldsymbol{\iota})} + \partial_{xz} V(x, z, \boldsymbol{\iota}) \right] (1 - T'(z)),$$

and, using the Slutsky equation, we can compute the compensated elasticity

$$\epsilon^c(z, \boldsymbol{\iota}) = \epsilon(z, \boldsymbol{\iota}) - \eta(z, \boldsymbol{\iota}) = \Delta^{-1} [\partial_x V(x, z, \boldsymbol{\iota})] \frac{1 - T'(z)}{z}.$$

## H Gender-based Taxes

[ FIGURE 4 ABOUT HERE ]

An important feature of the type of gender-based tax that we consider here is that its impact is heterogeneous across households with identical earnings,  $z$ , depending on the fraction  $\zeta_a(t)$  of their income that is earned by spouse  $a$ . Figures 4 and 5 display the impact of introducing a small tax,  $t > 0$ , on spouse  $a$ 's earnings on the aggregate disutility of effort for couples which, absent this gender specific tax, would have the same  $\omega$  and hence the same earnings.

[ FIGURE 5 ABOUT HERE ]

Regarding utility, the introduction of  $t$  also has a direct impact on utility since it reduces household consumption through  $\chi [1 - t\zeta_a(t)]^{1-\kappa}$ . If a revenue preserving reform is considered (i.e., if  $\chi$  is increased to compensate for the extra resources raised by  $t$ ), then for the couples for which consumption is increased, the welfare of spouse  $a$  necessarily increases.

[ FIGURE 9 ABOUT HERE ]

In Figure 9, we consider different values for spouse  $a$ 's productivity and show that the pattern remains. If there is no dissonance, as in the right panel of Figure 8, then the share of families that lose, according to the planner's metric, increases substantially. Next we consider how the presence of a constant returns to scale household production function affects our findings.

**Household Production** The apparently paradoxical finding that taxing  $a$  leads to an increase in her utility was derived under the assumption that there is no household production and the assumption that power, as captured by  $\alpha$ , is unaffected by policy. We explore the latter momentarily. For now, some considerations regarding home production are due. As we shall see, the existence of household production may reverse but may also strengthen the paradoxical findings of this section.

Assume that spouses have preferences defined over leisure,  $l$ , (price  $w_i$ ,  $i = a, b$ ), a market good,  $x$ , (price 1) and a household good (price  $\pi(w_a, w_b)$ , under the assumption that the household technology exhibits constant returns to scale). Associate to spouses the expenditure functions:

$$E^i(w_i, \pi, u_i) = \min_{x^m, x^d, l} x^m + \pi x^d + w_i l \quad \text{s.t.}, \quad U(x^m, x^d, l) \geq u_i$$

Then, the household cost minimization program,

$$\min E^a(w_a, \pi, u_a) + E^b(w_b, \pi, u_b)$$

s.t.  $\alpha u_a + (1 - \alpha)u_b \geq k$  allows us to write

$$du_a = - \left\{ \frac{\Delta_{w_a}}{\Delta} + \frac{\Delta_\pi}{\Delta} \pi_{w_a} \right\} dw_a, \quad (\text{H.1})$$

where,

$$\Delta_{w_a} = E_{uw_a}^a(w_a, \pi, u_a), \quad \Delta_\pi = E_{u\pi}^a(w_a, \pi, u_a) - \frac{\alpha}{1 - \alpha} E_{u\pi}^b(w_b, \pi, u_b),$$

and

$$\Delta = E_{uu}^a(w_a, \pi, u_a) - \left[ \frac{\alpha}{1 - \alpha} \right]^2 E_{uu}^b(w_b, \pi, u_b) > 0.$$

$E_{uw_a}^a(w_a, \pi, u_a)$  is always positive if leisure is a normal good. This is the impact that our model with separable structure and without household production captures. The last term is novel and is due to the presence of household production. It is not hard to see that

$$E_{u\pi}^i(w_i, \pi, u_i) = \frac{\partial x^d}{\partial u},$$

which is, in turn proportional to the income effect on spouse  $i$ 's demand for domestic good. This means that the sign of the second term, its sign depends on the difference between the share weighted income elasticity of demand for the household goods by the two spouses,  $\eta^a s^a - \eta^b s^b$  for  $s^i = \pi x_i^d / y_i$ , – and its strength on the participation of spouse  $a$ 's work on the production of  $x^d = x_a^d + x_b^d$ .

# **I Table and Figures**

Table 3: Inequality Measures Implied by the Optimal Tax

|               | Household Level | Individual Level | Household Level | Individual Level | Household Level | Individual Level |
|---------------|-----------------|------------------|-----------------|------------------|-----------------|------------------|
|               | $\alpha = .3$   |                  | $\alpha = .5$   |                  | $\alpha = .7$   |                  |
| No Dissonance | 0.1119          | 0.2560           | 0.1084          | 0.1084           | 0.1103          | 0.2552           |
| Dissonance    | 0.0517          | 0.2258           | 0.1084          | 0.1084           | 0.0894          | 0.2447           |

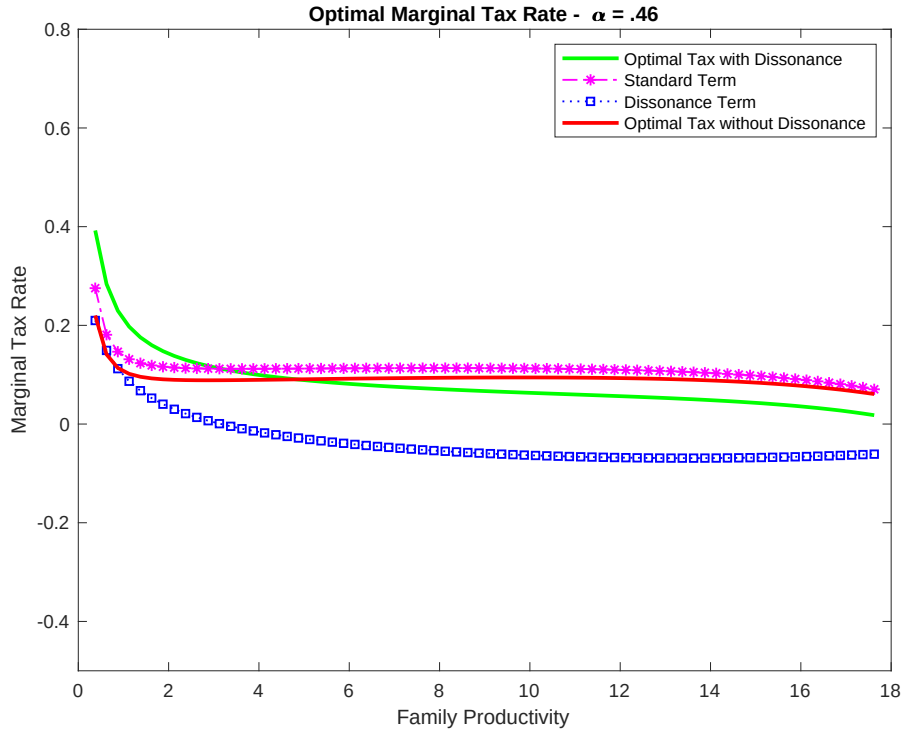


Figure 1: The figure simulates the shape of the optimal tax schedule calibrated for the US economy. The solid green line depicts the optimal marginal tax rate (MTR) as in Corollary ?? where the planner maximizes individual-oriented utilitarian criteria. The dotted lines decompose the MTR into a standard Mirrleesian term (pink with diamond markers), and a novel modified part (blue with square markers) induced from dissonance. The solid red line plots the optimal MTR as in Equation ?? where the planner maximizes household-oriented utilitarian criteria. These figures shed light on the impacts of families' dynamics in the optimality of policy decisions.

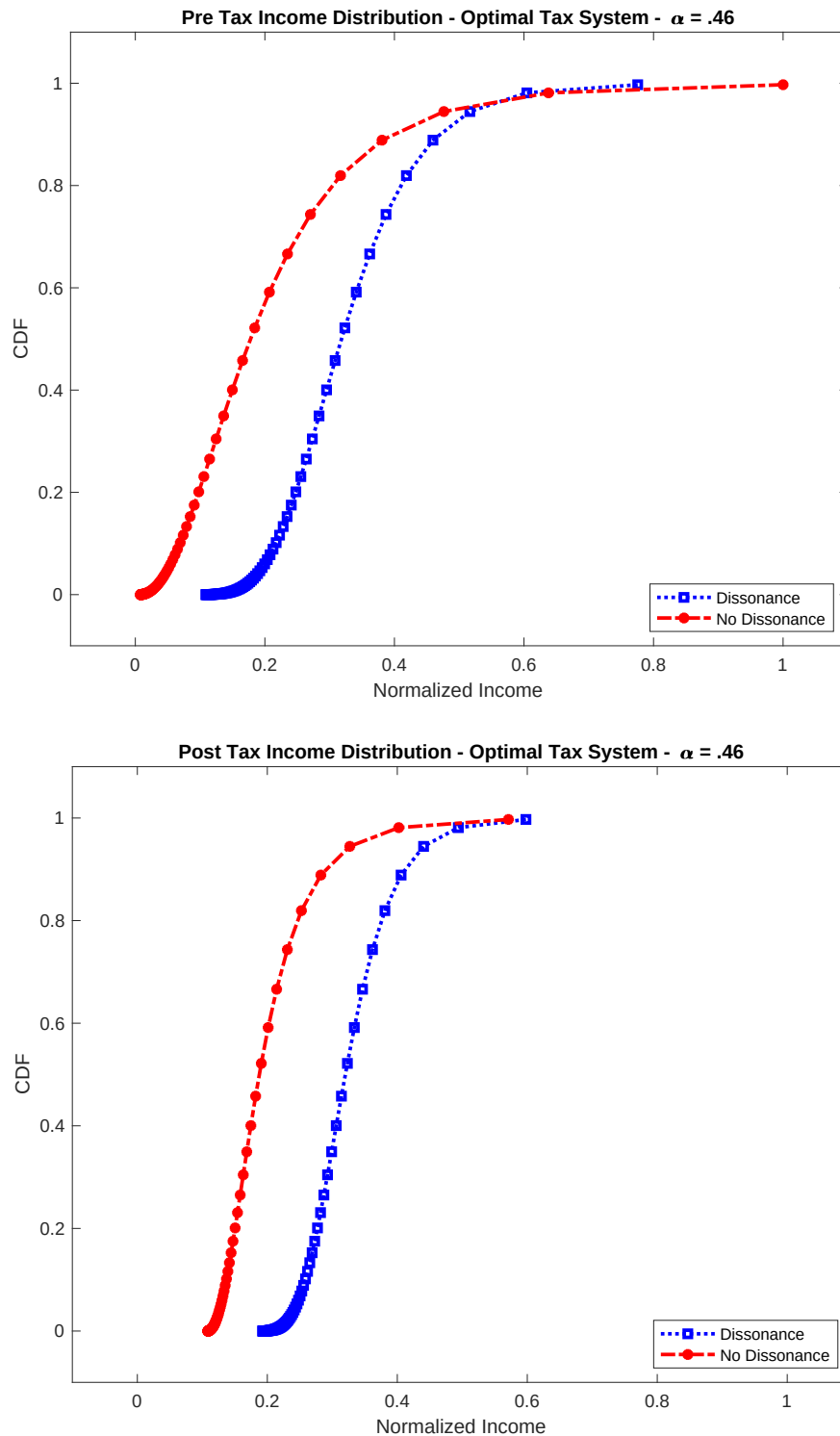


Figure 2: The figure displays the distribution of pre- and after tax household income for optimal taxes under a individual-based (Dissonance) and a household-based (No-dissonance) government objective. Graphs are normalized by the highest income attained under the household-based objective.

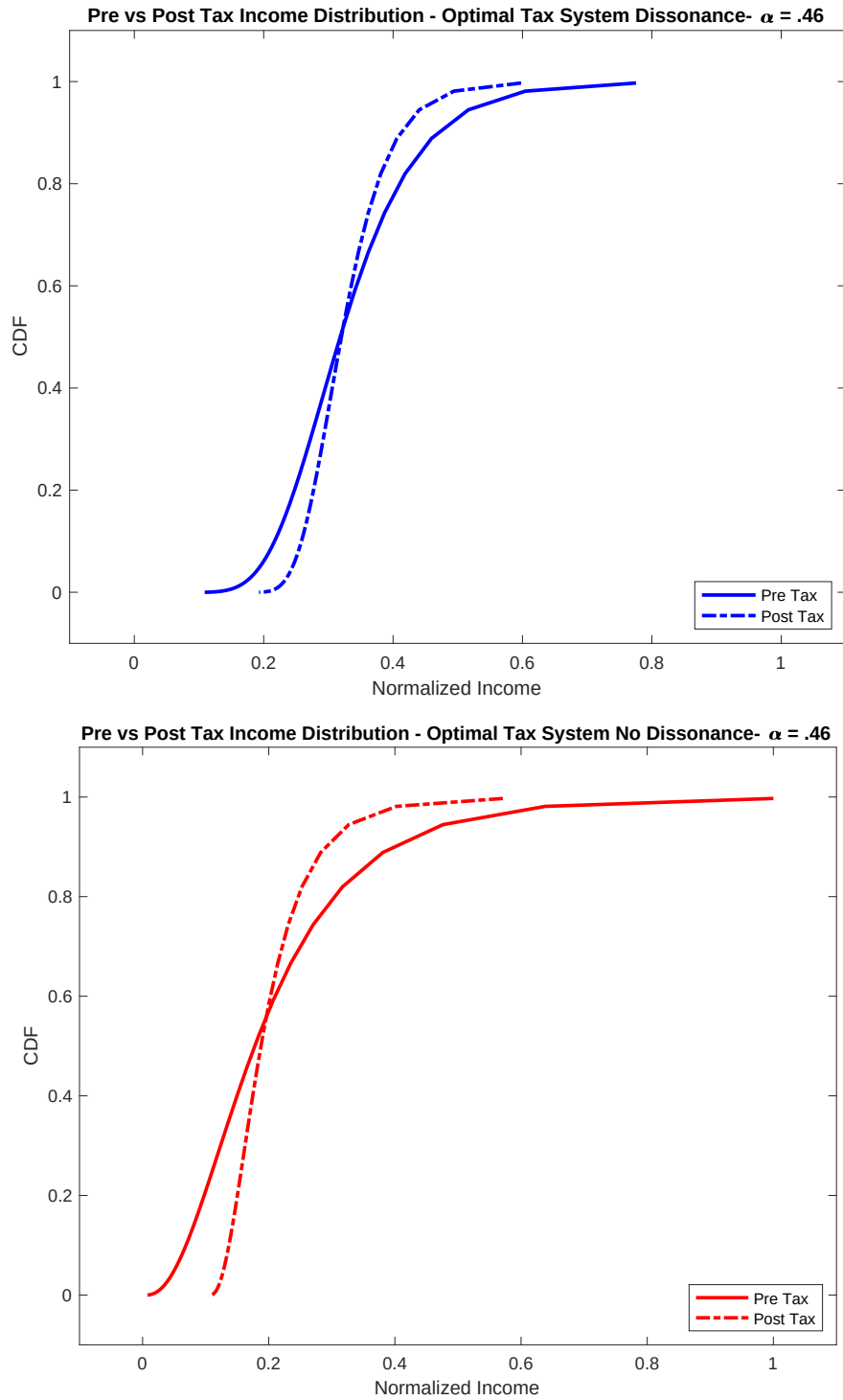


Figure 3: The figure displays the distribution of pre- and after tax household income for optimal taxes under a individual-based (Dissonance) and a household-based (No-dissonance) government objective. Graphs are normalized by the highest income attained under the household-based objective.

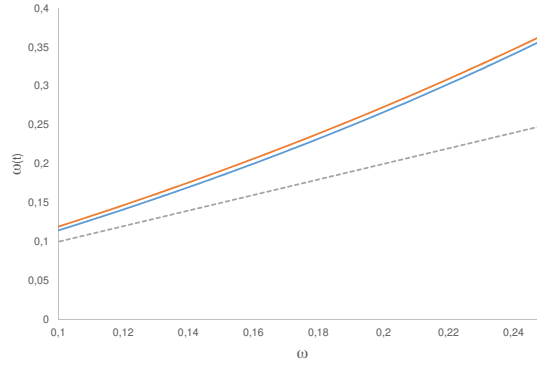


Figure 4: The figure shows how the introduction of a tax on spouse  $a$ 's earnings changes the household disutility of effort for couples with the same initial  $\omega$ . We consider two different values for  $\alpha$  and let  $\theta_b$  respond while holding  $\theta_a$  fixed to match the initial  $\omega$ . The gray curve (45° line) corresponds to  $\omega(0)$ , whereas the blue and red curves display  $\omega(t)$  for  $t > 0$  for the different combinations of  $\alpha$  and  $\theta_b$  that lead to the same  $\omega(0)$ .

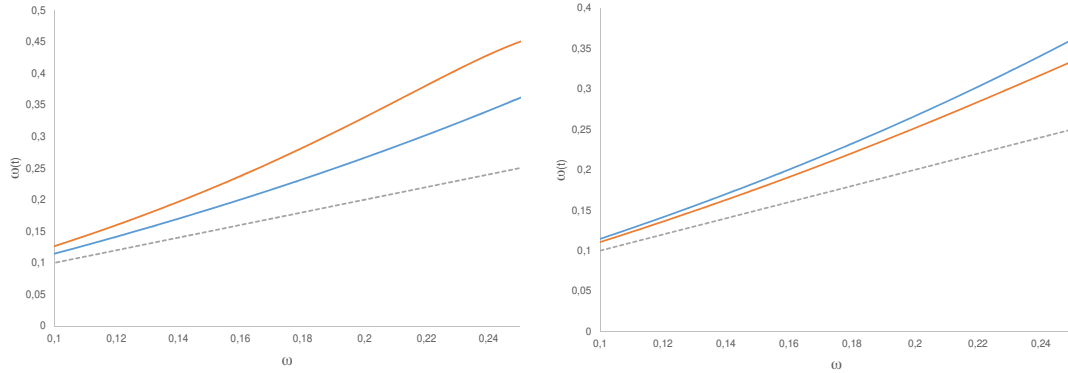


Figure 5: The figure shows how the introduction of a tax on spouse  $a$ 's earnings changes the household disutility of effort for couples with the same initial  $\omega$ . In the left panel, for every  $\omega$ , we consider different pairs of values for  $\theta_b$  and  $\alpha$  that lead to  $\omega$  keeping  $\theta_a$  fixed when  $t = 0$ . The blue and red curves show  $\omega(t)$  for the different combinations of  $\theta_b$  and  $\alpha$  when  $t > 0$ . In the right panel, for each  $\omega$ , we consider different pairs of  $\theta_a$  and  $\theta_b$  keeping  $\alpha$  fixed.



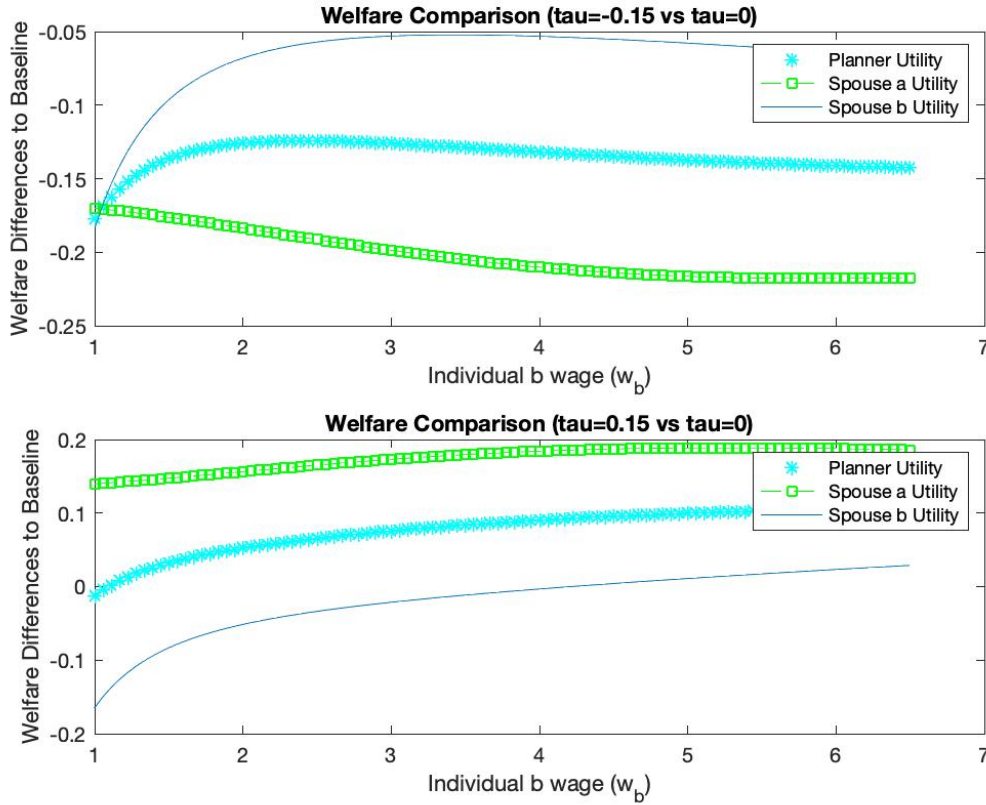


Figure 6: The figure shows welfare gains for spouse  $a$ , spouse  $b$ , and the social planner's own evaluation of household welfare change,  $DW = .5Dv_a + .5Dv_b$ , as a function of  $w_b = \theta_b^{-1/1+\gamma}$ . Introducing a small tax (bottom panel),  $\tau = 0.05$ , or subsidy (upper panel),  $\tau = -0.05$ , on  $a$ .  $w_a$  is held fixed at  $w_a = 3$  and  $\alpha = 0.3$ . Total revenues are held fixed by adjusting  $\chi$ . It is assumed that the distribution of  $w_b$  conditional on  $w$  is uniform.

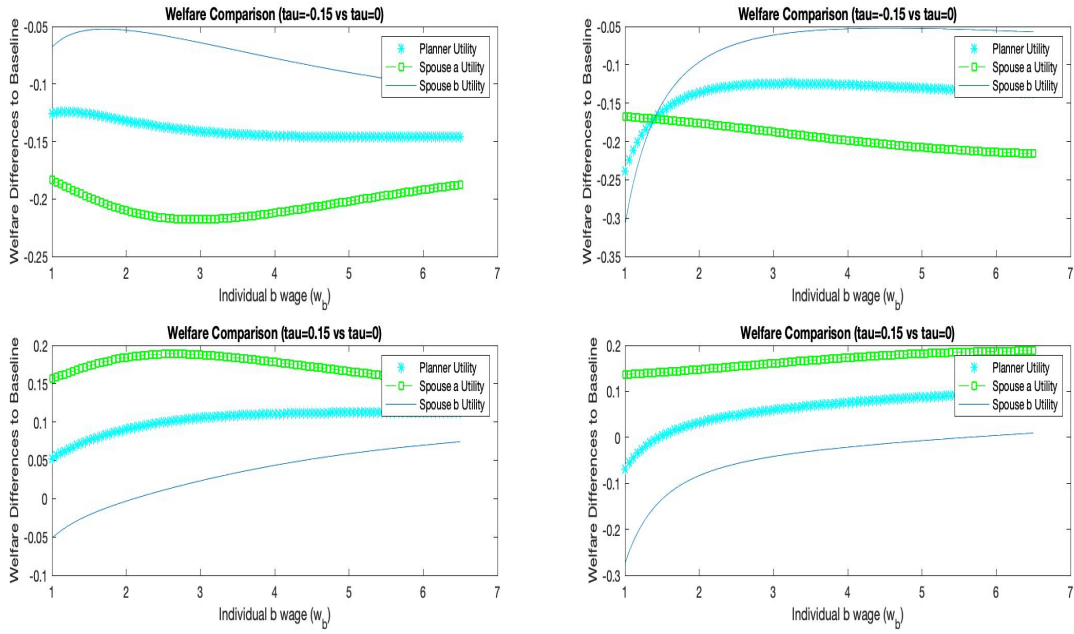


Figure 7: The only difference to Figure 8 is that  $w_a$  varies. On the left panel  $w_a$  is held fixed at  $w_a = 1.5$  and  $\alpha = 0.3$ . On the right panel  $w_a$  is held fixed at  $w_a = 4$  and  $\alpha = 0.3$ . The figure shows welfare gains for spouse  $a$ , spouse  $b$ , and the social planner's own evaluation of household welfare change,  $DW = .5Dv_a + .5Dv_b$ , as a function of  $w_b = \theta_b^{-1/1+\gamma}$ . Introducing a small tax (bottom panel),  $\tau = 0.05$ , or subsidy (upper panel),  $\tau = -0.05$  on  $a$ . Total revenues are held fixed by adjusting  $\chi$ . It is assumed that the distribution of  $w_b$  conditional on  $w$  is uniform.

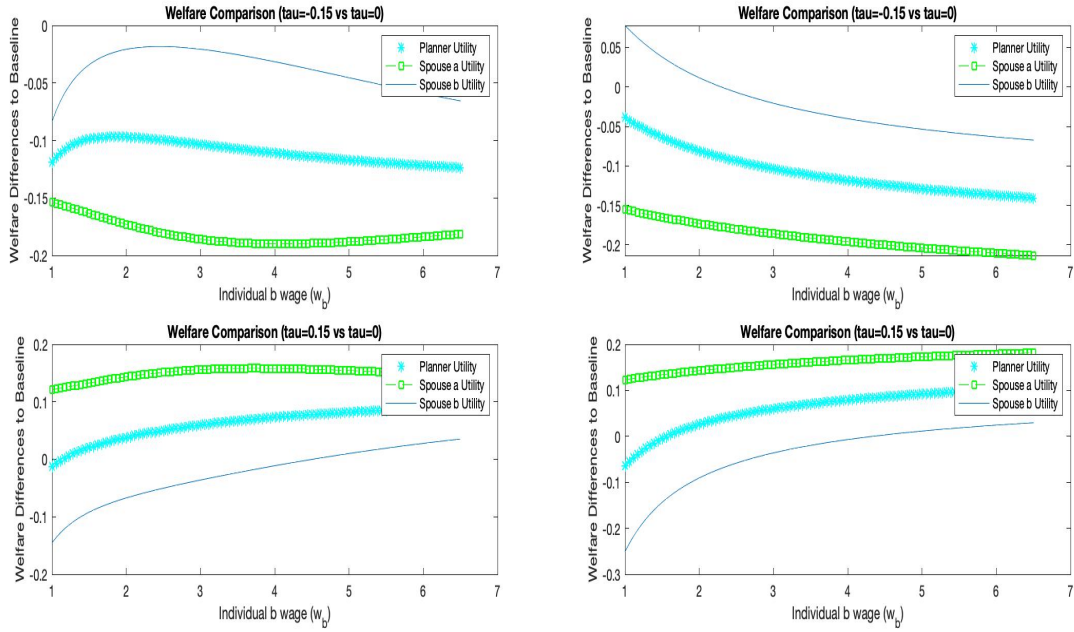


Figure 8: The only difference to Figure 9 is that  $\alpha$  varies. On the left panel  $w_a$  is held fixed at  $w_a = 3$  and  $\alpha = 0.5$ . On the right panel  $w_a$  is held fixed at  $w_a = 3$  and  $\alpha$  is endogenous to spouses' relative productivity ( $\frac{w_a}{w_a + w_b}$ ). The figure shows welfare gains for spouse  $a$ , spouse  $b$ , and the social planner's own evaluation of household welfare change,  $DW = .5Dv_a + .5Dv_b$ , as a function of  $w_b = \theta_b^{-1/1+\gamma}$ . Introducing a small tax (bottom panel),  $\tau = 0.05$ , or subsidy (upper panel),  $\tau = -0.05$  on  $a$ . Total revenues are held fixed by adjusting  $\chi$ . It is assumed that the distribution of  $w_b$  conditional on  $w$  is uniform.

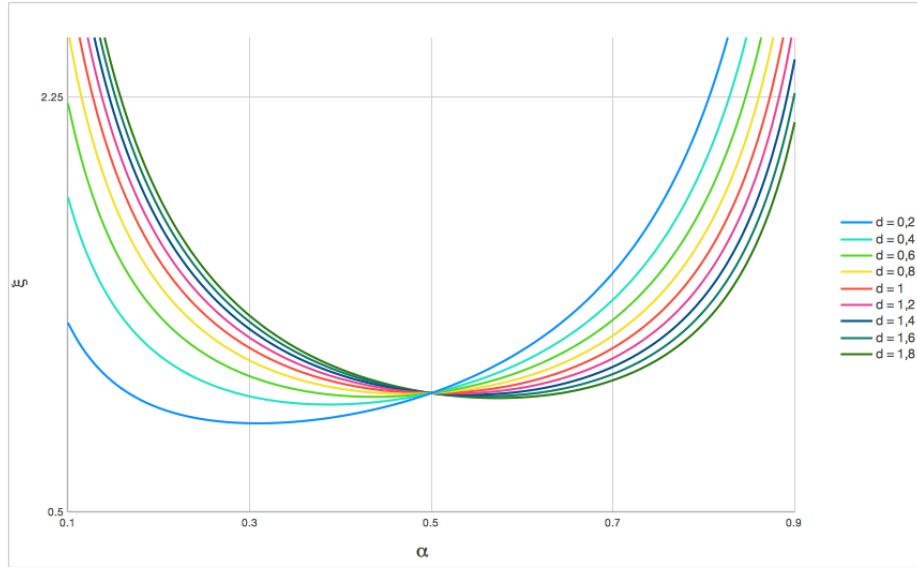


Figure 9: For different values of  $d = [w_a/w_b]^{1+\gamma}$  and preferences of the form studied in Section 3 the figure displays how the correction term  $\xi(z, \iota)$  varies with  $\alpha$ . For this specification,  $\xi(z, \iota)$  is independent of  $z$ .

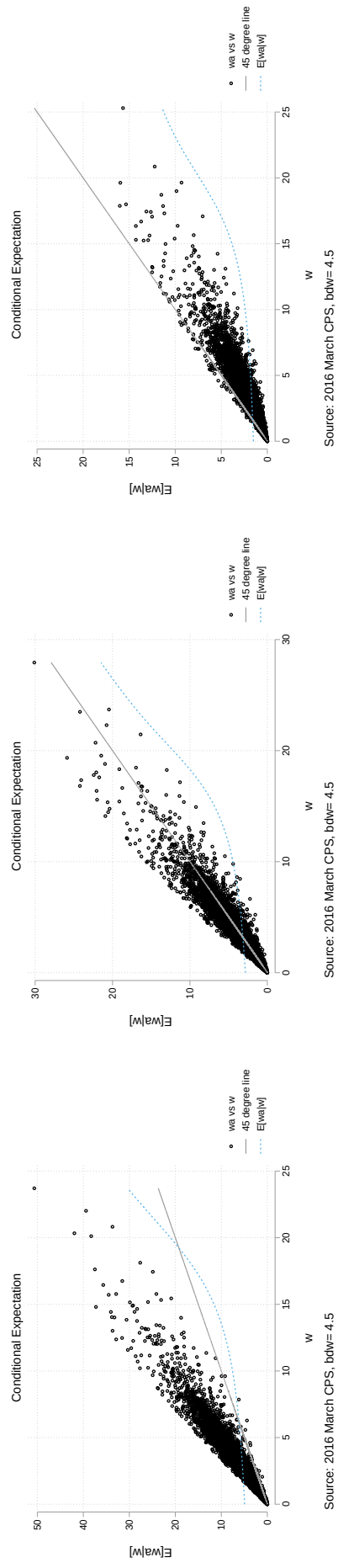
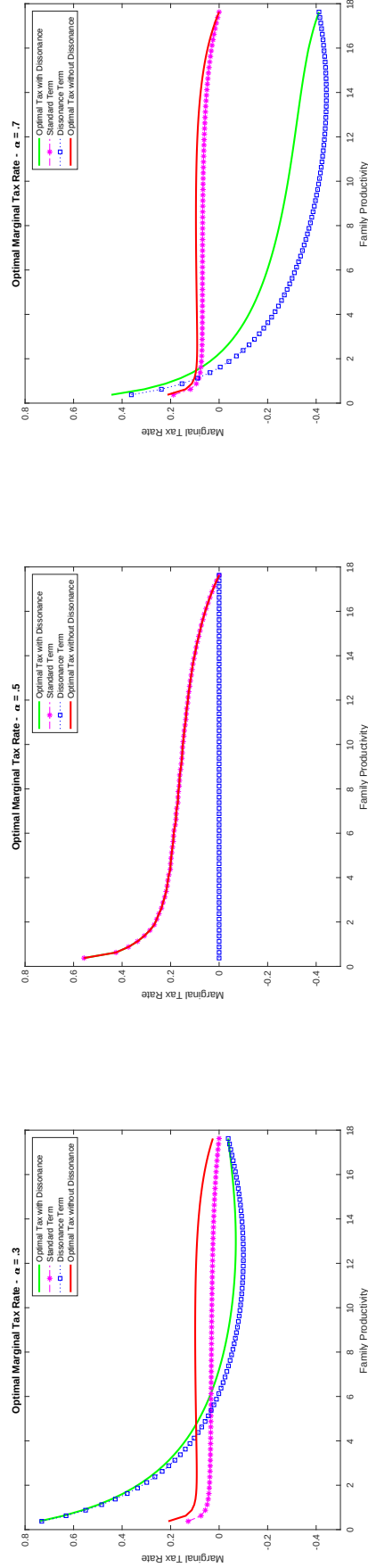


Figure 10: This figure shows the conditional expectation of spouse  $a$ 's modified type  $\omega_a$  given the household's aggregate type  $(\mathbb{E}[\omega_a|\theta])$ . This expectation varies with spouses income and Pareto weight ( $\alpha$ ). As  $\alpha$  is perturbed in each figure, we can see shifts in the conditional expectation. The plotted joint income comes from the empirical distribution based on the 2016 March CPS.



(a)  $\alpha = .3$

(b)  $\alpha = .5$

(c)  $\alpha = .7$

Figure 11: The figure simulates the shape of the optimal tax schedule calibrated for the US economy. The solid green line depicts the optimal marginal tax rate (MTR) as in Corollary ?? where the planner maximizes individual-oriented utilitarian criteria. The dotted lines decompose the MTR into a standard Mirrleesian term (pink with diamond markers), and a novel modified part (blue with square markers) induced from dissonance. The solid red line plots the optimal MTR as in Equation ?? where the planner maximizes household-oriented utilitarian criteria. These figures shed light on the impacts of families' dynamics in the optimality of policy decisions.

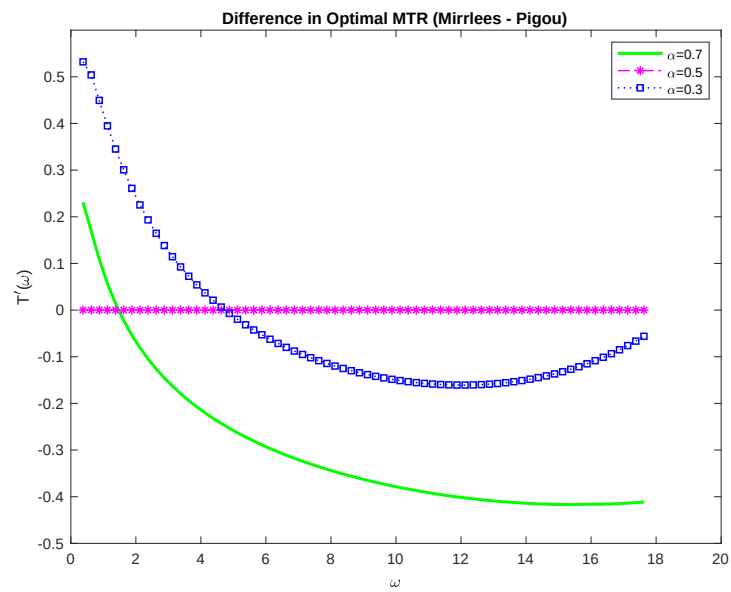


Figure 12: The figure plots the difference between the optimal marginal tax rate without dissonance (as in (??)) and the optimal marginal tax rate with the Pigouvian correction (as in Corollary ??). This difference varies with  $\alpha$  and  $\theta$ . When  $\alpha = 0.5$  this difference collapses to zero.

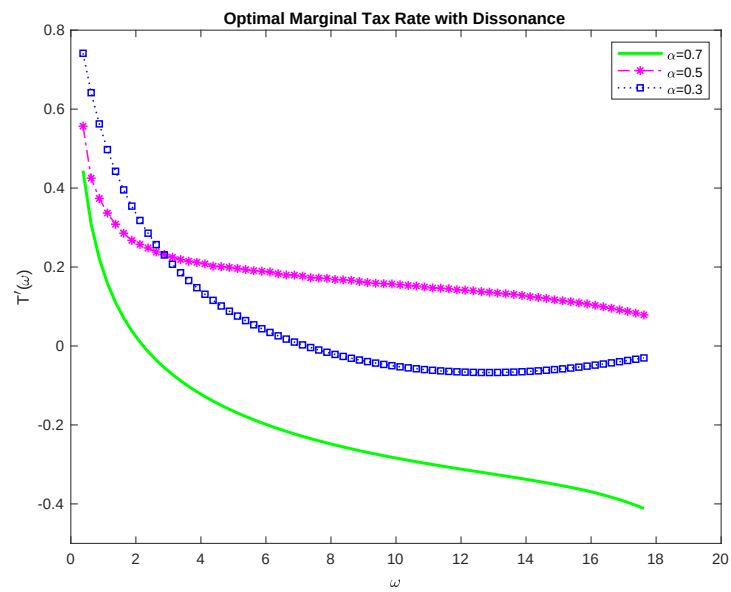


Figure 13: The figure shed light on the variation of optimal marginal tax rates when dissonance is accounted (as in Corollary ??), for different levels of the Pareto weight  $\alpha$ . This figure helps to compare the impact of optimal tax with dissonance across the productivity space.